

## HASH VERIFICATION

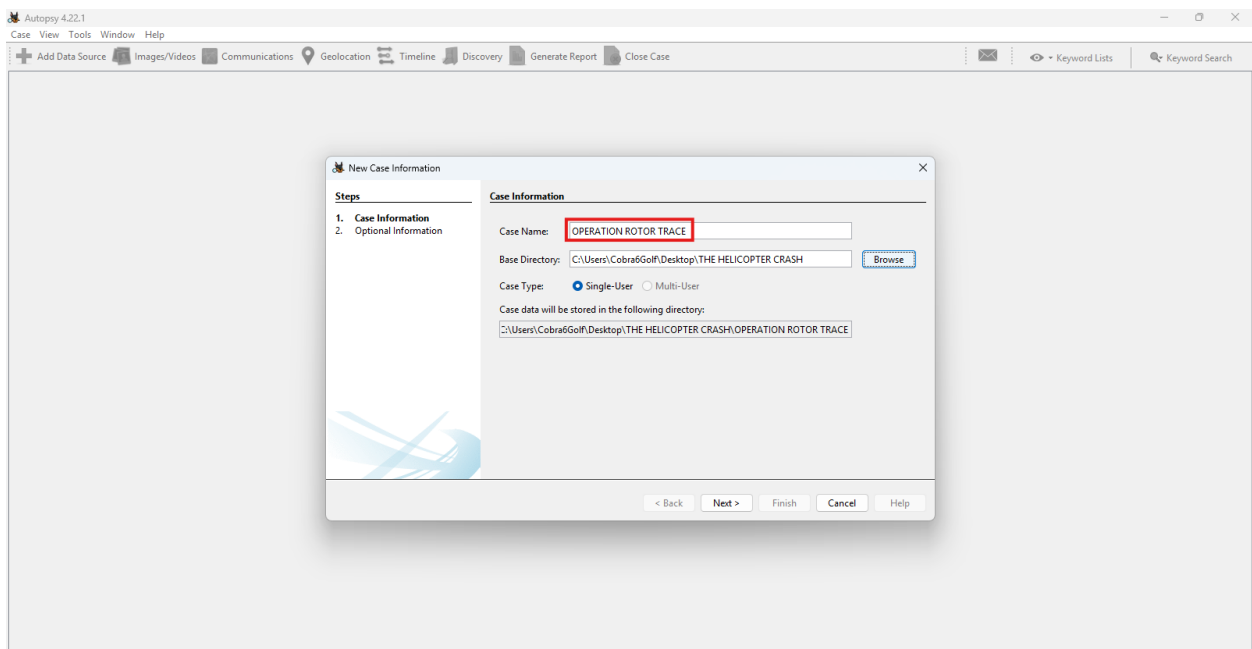
```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

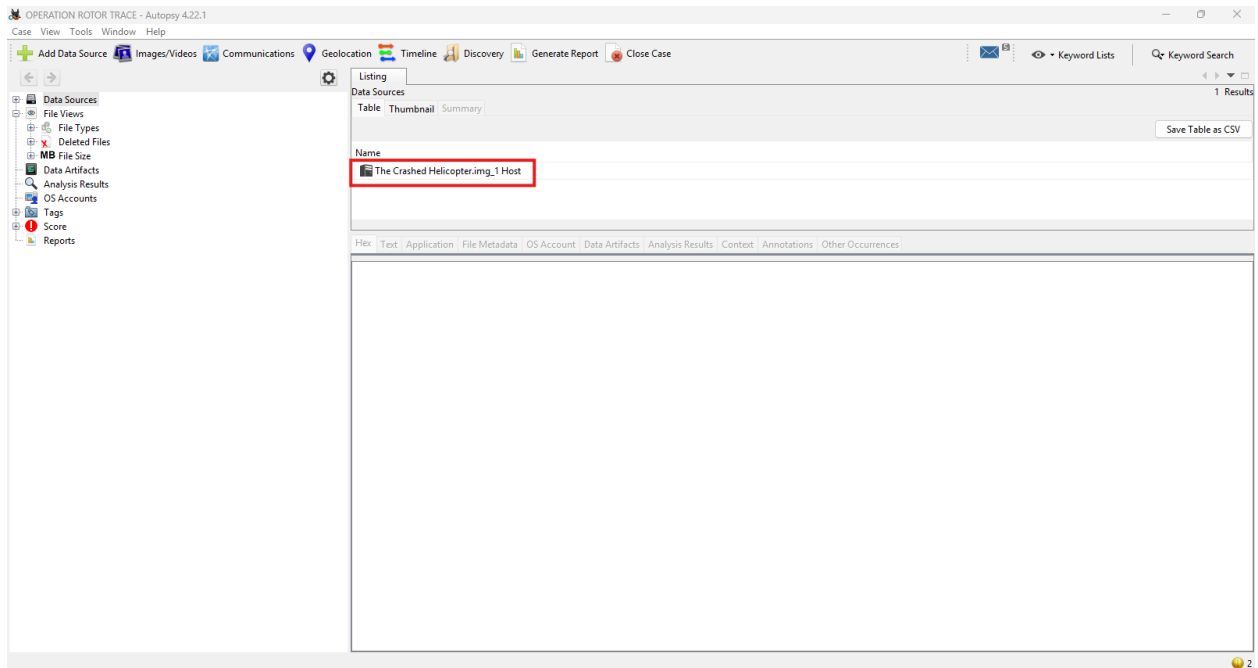
PS C:\WINDOWS\system32> Get-FileHash "C:\Users\Cobra6Golf\Desktop\THE HELICOPTER CRASH\The Crashed Helicopter*" -Algorithm SHA256
>>

Algorithm      Hash                                                    Path
-----
SHA256         3E98EB9913EAFB6157E4B37D1BC411788C73A183028CFC30B0EDDFE847A84E14  C:\Users\Cobra6Golf\Desktop\T...

PS C:\WINDOWS\system32> (Get-FileHash "C:\Users\Cobra6Golf\Desktop\THE HELICOPTER CRASH\The Crashed Helicopter*").Hash -eq "3e98eb9913eafb6157e4b37d1bc411788c73a183028cfc30b0eddf847a84e14"
True
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> (Get-FileHash "C:\Users\Cobra6Golf\Desktop\THE HELICOPTER CRASH\The Crashed Helicopter*" -Algorithm MD5).Hash -eq "00ff1dffa294cf10b769136bebbbeacf82"
True
PS C:\WINDOWS\system32>
```

## TOOL CASE INFORMATION





1. What was the original maintenance status of the Bell 412 before the flight, and what forensic evidence shows that the report was later changed to “CLEARED FOR FLIGHT”?

**Ans: The original maintenance status of the Bell 412 before the flight was "AIRCRAFT NOT SERVICEABLE." Forensic evidence shows the the maintenance audit was changed to CLEARED FOR FLIGHT.**

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$OrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unalloc (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - USB\_Backups (3)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)**
          - Temporary\_Review (8)
          - Downloads (5)
          - Pictures (3)
          - Maintenance (4)
          - Windows (4)
          - INF (4)
          - System32 (3)
          - config (5)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Users/kwame.adu/Documents

Table Thumbnail Summary

Save Table as CSV

| Name                                                   | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size  | Flags(Dir) | Known |
|--------------------------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|-------|------------|-------|
| [current folder]                                       |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096  | Allocated  | unkno |
| [parent folder]                                        |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096  | Allocated  | unkno |
| Temporary_Review                                       |   |   |   | 2026-08-17 22:02:46 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 22:02:46 UTC | 4096  | Allocated  | unkno |
| B412_Maintenance_Status_15Aug2026_FINAL.docx           |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52580 | Allocated  | unkno |
| <b>B412_Maintenance_Status_15Aug2026_ORIGINAL.docx</b> |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644 | Allocated  | unkno |
| Component_Certificate_XAL-412-7781.pdf                 |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 2404  | Allocated  | unkno |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text

XAVIER AIR LOGISTICS  
 BELL 412 MAINTENANCE STATUS - 15 AUGUST 2026  
**AIRCRAFT NOT SERVICEABLE**  
 AIRCRAFT  
 Bell 412 / Tail XAL-412  
 Defect  
 Intermittent gearbox oil-pressure indication observed during ground run. HUMS review shows an elevated gearbox-temperature trend compared with previous flights.  
 Engineering Assessment  
 Aircraft is not to be released until oil-pressure indication, sensor installation and component certification are verified.  
 Recommended Action  
 Inspect gearbox oil system, confirm installed sensor serial against certificate, repeat ground run and review HUMS trend before release.  
 CREATED BY HIVE CONSULT, NANA SEI ANYEMEDU | PROPERTY OF HIVE CONSULT  
 image1.png

-----METADATA-----

Content-Type: application/vnd.openxmlformats-officedocument.wordprocessingml.document  
 X-TIKA-Parsed-By: org.apache.tika.parser.DefaultParser  
 X-TIKA-Parsed-By-Full-Set: org.apache.tika.parser.DefaultParser  
 cprevision: 1

2

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$OrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unalloc (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - USB\_Backups (3)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)**
          - Temporary\_Review (8)
          - Downloads (5)
          - Pictures (3)
          - Maintenance (4)
          - Windows (4)
          - INF (4)
          - System32 (3)
          - config (5)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Users/kwame.adu/Documents

Table Thumbnail Summary

Save Table as CSV

| Name                                                | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size  | Flags(Dir) | Known |
|-----------------------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|-------|------------|-------|
| B412_Maintenance_Status_15Aug2026_FINAL.docx        |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52580 | Allocated  | unkno |
| B412_Maintenance_Status_15Aug2026_ORIGINAL.docx     |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644 | Allocated  | unkno |
| Component_Certificate_XAL-412-7781.pdf              |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 2404  | Allocated  | unkno |
| Diagnostic_Export_17Aug2026.csv                     |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 298   | Allocated  | unkno |
| <b>Maintenance_Component_and_Release_Audit.xlsx</b> |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 7138  | Allocated  | unkno |
| Supplier_Invoice_XAL-412-7781.pdf                   |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 2340  | Allocated  | unkno |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text

Audit Log

| Event UTC           | User      | Action      | Record Type    | Record ID | Old Value      | New Value                                       | Source Host                                    | Review                          |
|---------------------|-----------|-------------|----------------|-----------|----------------|-------------------------------------------------|------------------------------------------------|---------------------------------|
| 2026-08-17 20:18:41 | kwame.adu | EXPORT      | diagnostic_log |           |                | HUMS-0815                                       |                                                | Diagnostic_Export_15Aug2026.csv |
| 2026-08-17 21:38:52 | kwame.adu | OPEN        | document       |           | maint-status   | B412_Maintenance_Status_15Aug2026_ORIGINAL.docx | XAL-MAINT-07                                   | XAL-MAINT-07                    |
| 2026-08-17 21:43:49 | kwame.adu | EDIT        | document       |           | release_status | XAL-412                                         | AIRCRAFT NOT SERVICEABLE                       | CLEARED FOR FLIGHT              |
| 2026-08-17 21:44:26 | kwame.adu | SAVE_AS     | document       |           | maint-status   | B412_Maintenance_Status_15Aug2026_ORIGINAL.docx | B412_Maintenance_Status_15Aug2026_FINAL.docx   | XAL-MAINT-07                    |
| 2026-08-17 21:53:11 | kwame.adu | COPY_TO_USB | file           |           | maint-status   |                                                 | E\B412_Maintenance_Status_15Aug2026_FINAL.docx | XAL-MAINT-07                    |
| 2026-08-17 22:02:47 | kwame.adu | DELETE      | diagnostic_log |           |                | HUMS-0815                                       | Diagnostic_Export_15Aug2026.csv                | DELETED                         |

-----METADATA-----

Content-Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet  
 X-TIKA-Parsed-By: org.apache.tika.parser.DefaultParser  
 extended-properties: DocSecurityString: None  
 protected: false

2

- Who modified the maintenance report, when was it modified, and which digital artifacts can be used to attribute the change to a particular user or device?

**Ans: Kwame Adu modified the maintenance report on 17 August 2026 at 9:43:49 PM. The maintenance audit logs recorded that the release status of aircraft XAL-412 was changed from "AIRCRAFT NOT SERVICEABLE" to "CLEARED FOR FLIGHT."**

***The maintenance report was modified by Kwame Adu on 17 August 2026 at 21:43:49 UTC, as recorded in the maintenance audit logs. Attribution of the modification is supported by the presence of the file Maintenance\_Component\_and\_Release\_Audit.xlsx within the user profile directory C:\Users\kwame.adu\Documents\, indicating that the document was associated with the kwame.adu account. Additional supporting artifacts include the account SID S-1-5-21-20260818-1001, account information identifying the user as a Local Administrator, and user profile artifacts showing the account was active on the system. The maintenance audit log remains the primary evidence because it records the exact change from "AIRCRAFT NOT SERVICEABLE" to "CLEARED FOR FLIGHT", along with the responsible user and timestamp.***

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img\_The Crashed Helicopter.img/vol2/Windows/System32/config

| Name               | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(Meta) | Local |
|--------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-------------|-------|
| [current folder]   |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096 | Allocated  | unknown | Allocated   | /img  |
| [parent folder]    |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096 | Allocated  | unknown | Allocated   | /img  |
| Power_Events.csv   |   |   |   | 2026-08-18 07:51:28 UTC | 0000-00-00 00:00:00 | 2026-08-18 07:51:28 UTC | 2026-08-18 07:51:28 UTC | 395  | Allocated  | unknown | Allocated   | /img  |
| SYSTEM_PROFILE.txt |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 212  | Allocated  | unknown | Allocated   | /img  |
| User_Accounts.csv  |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 334  | Allocated  | unknown | Allocated   | /img  |

Save Table as CSV

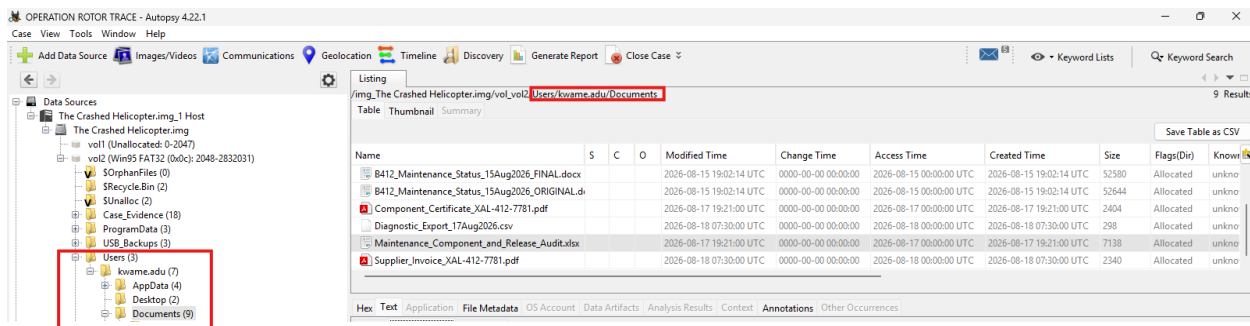
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset

Text Source: File Text

username.sid.account\_type.profile\_path.last\_logon\_off.status  
 kvame.edu.S-1-5-21-20260818-1001 Local Administrator C:\Users\kvame.edu.2026-08-18T07:32:41Z Active  
 engineering.admin.S-1-5-21-20260818-1002 Administrator C:\Users\engineering.admin.2026-08-17T12:148:02Z Active  
 Guest.S-1-5-21-20260818-501 Guest C:\Users\Guest\_Disabled



- What serial number appears on the replacement component installed on the helicopter, and does it match the component certificate, invoice and maintenance records?

**Ans: Examination of the maintenance records, installed component photograph, component certificate, and supplier invoice revealed a serial number discrepancy. The installed Gearbox Oil-Pressure Sensor Assembly carried serial number B412-OPS-883419, while both the component certificate and supplier invoice referenced serial number B412-OPS-883941. The maintenance register flagged this discrepancy with a status of "SERIAL MISMATCH", indicating that the component installed on aircraft XAL-412 was not the same component documented in the supporting certification and procurement records.**

| Source                                          | Serial Number   |
|-------------------------------------------------|-----------------|
| Installed Component Photograph (Visible Serial) | B412-OPS-883419 |
| Maintenance Record (Installed Serial)           | B412-OPS-883419 |
| Component Certificate (Certificate Serial)      | B412-OPS-883941 |
| Supplier Invoice (Serial Number Shipped)        | B412-OPS-883941 |

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img\_The Crashed Helicopter.img/vol2/Users/kwame.adu/Pictures/Maintenance

Table Thumbnail Summary

Save Table as CSV

| Name                                   | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size   | Flags(Dir) | Known   | File |
|----------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|--------|------------|---------|------|
| [current folder]                       |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096   | Allocated  | unknown | All  |
| [parent folder]                        |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096   | Allocated  | unknown | All  |
| Installed_Component_Serial.jpg         |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 156455 | Allocated  | unknown | All  |
| Installed_Component_Serial_Closeup.png |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 44739  | Allocated  | unknown | All  |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

0° 39% Reset

2026-08-17 18:48:09 UTC

### INSTALLED COMPONENT - XAL-412

GEARBOX OIL PRESSURE SENSOR

VISIBLE SERIAL  
B412-OPS-883419

CERTIFICATE SERIAL  
B412-OPS-883941

CAPTURED  
17 AUG 2026 18:48 UTC

SOURCE  
Maintenance Engineer Phone

CREATED BY HIVE CONSULT HANA SEI ANYEMEDU  
PRODUCED AT HIVE CONSULT

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img\_The Crashed Helicopter.img/vol2/Users/kwame.adu/Pictures/Maintenance

Table Thumbnail Summary

Save Table as CSV

| Name                                   | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size   | Flags(Dir) | Known   | File |
|----------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|--------|------------|---------|------|
| [current folder]                       |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096   | Allocated  | unknown | All  |
| [parent folder]                        |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096   | Allocated  | unknown | All  |
| Installed_Component_Serial.jpg         |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 156455 | Allocated  | unknown | All  |
| Installed_Component_Serial_Closeup.png |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 44739  | Allocated  | unknown | All  |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

0° 39% Reset

INSTALLED COMPONENT PHOTOGRAPH - SYNTHETIC TRAINING EVIDENCE

XAL-412 SENSOR  
B412-OPS-883419  
INSTALLED 15 AUG 2026

CREATED BY HIVE CONSULT HANA SEI ANYEMEDU  
PRODUCED AT HIVE CONSULT

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$OrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unlloc (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - USB\_Backups (3)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)
            - Temporary\_Review (8)
            - Downloads (5)
            - Pictures (3)
            - Maintenance (4)
          - Windows (4)
          - INF (4)
          - System32 (3)
          - config (5)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Users/kwame.adu/Documents

Table Thumbnail Summary

Save Table as CSV

| Name                                            | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size  | Flags(Dir) | Known   |
|-------------------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|-------|------------|---------|
| Temporary_Review                                |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52580 | Allocated  | unknown |
| B412_Maintenance_Status_15Aug2026_FINAL.docx    |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644 | Allocated  | unknown |
| B412_Maintenance_Status_15Aug2026_ORIGINAL.docx |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644 | Allocated  | unknown |
| Component_Certificate_XAL-412-7781.pdf          |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 2404  | Allocated  | unknown |
| Diagnostic_Export_17Aug2026.csv                 |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 298   | Allocated  | unknown |
| Maintenance_Component_and_Release_Audit.xlsx    |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 7138  | Allocated  | unknown |
| Supplier_Invoice_XAL-412-7781.pdf               |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 2340  | Allocated  | unknown |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Content Annotations Other Occurrences

COMPONENT CERTIFICATE

Component

B412 Gearbox Oil-Pressure Sensor Assembly

Certificate Serial

B412-OPS-683941

Supplier

AeroWest Components Ltd (fictional)

Certificate Status

Training certificate - synthetic

Remarks

Serial on this certificate must match the installed component before release to service.

Page 1 / 1

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$OrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unlloc (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - USB\_Backups (3)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)
            - Temporary\_Review (8)
            - Downloads (5)
            - Pictures (3)
            - Maintenance (4)
          - Windows (4)
          - INF (4)
          - System32 (3)
          - config (5)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Users/kwame.adu/Documents

Table Thumbnail Summary

Save Table as CSV

| Name                                            | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size  | Flags(Dir) | Known   |
|-------------------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|-------|------------|---------|
| Temporary_Review                                |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52580 | Allocated  | unknown |
| B412_Maintenance_Status_15Aug2026_FINAL.docx    |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644 | Allocated  | unknown |
| B412_Maintenance_Status_15Aug2026_ORIGINAL.docx |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644 | Allocated  | unknown |
| Component_Certificate_XAL-412-7781.pdf          |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 2404  | Allocated  | unknown |
| Diagnostic_Export_17Aug2026.csv                 |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 298   | Allocated  | unknown |
| Maintenance_Component_and_Release_Audit.xlsx    |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 7138  | Allocated  | unknown |
| Supplier_Invoice_XAL-412-7781.pdf               |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 2340  | Allocated  | unknown |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Content Annotations Other Occurrences

SUPPLIER INVOICE

Invoice

AWC-2026-7781

Component

Gearbox Oil-Pressure Sensor Assembly

Serial Number Shipped

B412-OPS-683941

Amount

GHS 48,750.00

Delivery

Xavier Air Logistics Engineering Stores - 14 Aug 2026

Page 1 / 1

| Record  | Aircraft | Component                            | Certificate Serial | Installed Serial | Installed UTC       | Engineer         | Status          |
|---------|----------|--------------------------------------|--------------------|------------------|---------------------|------------------|-----------------|
| CR-7781 | XAL-412  | Gearbox Oil-Pressure Sensor Assembly | B412-OPS-883941    | B412-OPS-883419  | 2026-08-15 17:42:11 | Kwame Adu        | SERIAL MISMATCH |
| CR-7772 | XAL-412  | Hydraulic Filter                     | HYD-412-771042     | HYD-412-771042   | 2026-07-21 11:08:52 | Emmanuel Quartey | VERIFIED        |
| CR-7760 | XAL-412  | Rotor Track Sensor                   | RTS-412-550811     | RTS-412-550811   | 2026-06-30 09:30:15 | Kwame Adu        | VERIFIED        |

- What suspicious browser searches were performed on the maintenance laptop, and what do their timestamps reveal about possible attempts to delete or conceal maintenance evidence?

**Ans: Analysis of the Chrome browser history revealed searches related to deleting logs, removing metadata, clearing browser history, and removing recent file traces. These searches are indicative of attempts to conceal user activity. Notably, searches for "remove recent files windows 11" and "clear browser history permanently" occurred approximately 14 minutes and 18 minutes respectively after the maintenance record was changed from "AIRCRAFT NOT SERVICEABLE" to "CLEARED FOR FLIGHT." This timing is consistent with a possible attempt to conceal evidence of the maintenance record modification and associated user activity.**

| Search                                                      | Last Visit Time   | Last Visit Time (UTC) |
|-------------------------------------------------------------|-------------------|-----------------------|
| Search - Bell 412 gearbox warning code                      | 13431292931000000 | 2026-08-15 18:42:11   |
| Search - oil pressure sensor Bell 412                       | 13431293049000000 | 2026-08-15 18:44:09   |
| Search - aircraft HUMS log storage location                 | 13431471513000000 | 2026-08-17 20:18:33   |
| Search - how to clear aircraft maintenance software history | 13431471737000000 | 2026-08-17 20:22:17   |
| Search - delete diagnostic logs windows                     | 13431471898000000 | 2026-08-17 20:24:58   |
| Search - remove metadata from word document                 | 13431472184000000 | 2026-08-17 20:29:44   |
| Search - USB connection history windows                     | 13431472272000000 | 2026-08-17 20:31:12   |
| Search - remove recent files windows 11                     | 13431477487000000 | 2026-08-17 21:58:07   |
| Search - clear browser history permanently                  | 13431477701000000 | 2026-08-17 22:01:41   |

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img\_The Crashed Helicopter.img/vol\_vo2/Users/kwame.adu/AppData/Local/Google/Chrome/User Data/Default

Table Thumbnail Summary

Save Table as CSV

| Name             | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size   | Flags(Dir) | Known   | Flags(Meta) | Location    |
|------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|--------|------------|---------|-------------|-------------|
| [current folder] |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096   | Allocated  | unknown | Allocated   | /img_The Cr |
| [parent folder]  |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096   | Allocated  | unknown | Allocated   | /img_The Cr |
| History          |   |   |   | 2026-08-17 22:01:40 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 22:01:40 UTC | 196608 | Allocated  | unknown | Allocated   | /img_The Cr |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table urls 1800 entries Page 1 of 19 Export to CSV

| id | url                                                                          | title                                                       | visit_cou... | typed_c... | last_visit_time   | hidden |
|----|------------------------------------------------------------------------------|-------------------------------------------------------------|--------------|------------|-------------------|--------|
| 1  | https://search.example/?q=Bell+412+gearbox+warning+code                      | Search - Bell 412 gearbox warning code                      | 1            | 0          | 13431292931000000 | 0      |
| 2  | https://search.example/?q=oil+pressure+sensor+Bell+412                       | Search - oil pressure sensor Bell 412                       | 1            | 0          | 13431293049000000 | 0      |
| 3  | https://search.example/?q=aircraft+HUMS+log+storage+location                 | Search - aircraft HUMS log storage location                 | 1            | 0          | 13431471513000000 | 0      |
| 4  | https://search.example/?q=how+to+clear+aircraft+maintenance+software+history | Search - how to clear aircraft maintenance software history | 1            | 0          | 13431471737000000 | 0      |
| 5  | https://search.example/?q=delete+diagnostic+logs+windows                     | Search - delete diagnostic logs windows                     | 1            | 0          | 13431471898000000 | 0      |
| 6  | https://search.example/?q=remove+metadata+from+word+document                 | Search - remove metadata from word document                 | 1            | 0          | 13431472184000000 | 0      |
| 7  | https://search.example/?q=USB+connection+history+windows                     | Search - USB connection history windows                     | 1            | 0          | 13431472272000000 | 0      |
| 8  | https://search.example/?q=remove+recent+files+windows+11                     | Search - remove recent files windows 11                     | 1            | 0          | 13431477487000000 | 0      |
| 9  | https://search.example/?q=clear+browser+history+permanently                  | Search - clear browser history permanently                  | 1            | 0          | 13431477701000000 | 0      |
| 10 | https://supplier.example/components/                                         | Weather Briefing                                            | 1            | 0          | 13429017720000000 | 0      |
| 11 | https://search.example/?q=                                                   | Xavier Air Maintenance Portal                               | 1            | 0          | 13429018800000000 | 0      |
| 12 | https://docs.xavierair.example/maintenance/                                  | Aviation Manual                                             | 1            | 0          | 13429019880000000 | 0      |
| 13 | https://mail.xavierair.example/                                              | Webmail                                                     | 1            | 0          | 13429020960000000 | 0      |
| 14 | https://search.example/?q=                                                   | News                                                        | 1            | 0          | 13429022040000000 | 0      |
| 15 | https://docs.xavierair.example/maintenance/                                  | Weather Briefing                                            | 1            | 0          | 13429023120000000 | 0      |
| 16 | https://supplier.example/components/                                         | Aviation Manual                                             | 1            | 0          | 13429024200000000 | 0      |
| 17 | https://mail.xavierair.example/                                              | News                                                        | 1            | 0          | 13429025280000000 | 0      |
| 18 | https://news.example/                                                        | Maps                                                        | 1            | 0          | 13429026360000000 | 0      |

## Evidence Corroborating when compared with the maintenance audit log:

| Event                                                                          | Time (UTC)          |
|--------------------------------------------------------------------------------|---------------------|
| Maintenance status changed from AIRCRAFT NOT SERVICEABLE to CLEARED FOR FLIGHT | 2026-08-17 21:43:49 |
| Search - remove recent files windows 11                                        | 2026-08-17 21:58:07 |
| Search - clear browser history permanently                                     | 2026-08-17 22:01:41 |

**Analysis of the Chrome browser history revealed several searches related to the removal of digital traces and system activity records. Searches for "how to clear aircraft maintenance software history," "delete diagnostic logs windows," "remove metadata from word document," and "USB connection history windows" were conducted prior to the modification of the aircraft maintenance status. These searches indicate an interest in methods for removing or obscuring forensic evidence from the system. Further examination showed that searches for "remove recent files windows 11" at 21:58:07 UTC and "clear browser history permanently" at 22:01:41 UTC occurred after the maintenance record was altered from "AIRCRAFT NOT SERVICEABLE" to "CLEARED FOR FLIGHT" at 21:43:49 UTC. These browser activities constitute corroborating evidence and are temporally consistent with a potential attempt to conceal or remove traces of the maintenance record modification and associated user activity.**

5. What does the recovered USB evidence reveal about the original maintenance report, diagnostic exports, component documentation and deleted files, and when was the USB connected to the maintenance laptop?

***Ans: Examination of the recovered USB device identified copies of the maintenance report, component certificate, supplier invoice, and installed component photographs relating to aircraft XAL-412. In addition, several deleted files were recovered from the maintenance laptop, including a document titled B412\_NOT\_SERVICEABLE.pdf, a diagnostic export, gearbox inspection notes, and a maintenance history backup database. These artifacts suggest that original maintenance and diagnostic records existed prior to later modifications and deletion attempts. USB connection records further established that the Kingston DataTraveler 3.0 device labelled XAL\_MAINT\_16GB was first connected on 15 August 2026 at 18:14:08 UTC and last connected on 17 August 2026 at 22:03:12 UTC by user kwame.adu, shortly after the maintenance status was changed and after browser searches related to deleting evidence were performed. This timing provides corroborating evidence linking the USB activity to the maintenance record alterations.***

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - SOrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unalloc (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - USB\_Backups (3)
      - XAL\_MAINT\_16GB (8)**
      - Users (3)
      - Windows (4)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/USB\_Backups/XAL\_MAINT\_16GB

Table Thumbnail Summary

Save Table as CSV

| Name                                            | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size   | Flags(Dir) | Known   |
|-------------------------------------------------|---|---|-------------------------|---------------------|-------------------------|-------------------------|--------|------------|---------|
| [current folder]                                |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 4096   | Allocated  | unknown |
| [parent folder]                                 |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 4096   | Allocated  | unknown |
| B412_Maintenance_Status_15Aug2026_FINAL.docx    |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52580  | Allocated  | unknown |
| B412_Maintenance_Status_15Aug2026_ORIGINAL.docx |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 52644  | Allocated  | unknown |
| Component_Certificate_XAL-412-7781.pdf          |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 2404   | Allocated  | unknown |
| Installed_Component_Serial.jpg                  |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 156455 | Allocated  | unknown |
| Installed_Component_Serial_Closeup.png          |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 44739  | Allocated  | unknown |
| Supplier_Invoice_XAL-412-7781.pdf               |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 2340   | Allocated  | unknown |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - SOrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unalloc (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - XavierAir (3)
      - USB\_Backups (3)
      - XAL\_MAINT\_16GB (8)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)**
            - Temporary\_Review (8)**
          - Downloads (5)
          - Pictures (3)
        - Windows (4)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Users/kwame.adu/Documents/Temporary\_Review

Table Thumbnail Summary

Save Table as CSV

| Name                               | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size  | Flags(Dir)  | Known   | Flags(Dir)  |
|------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|-------|-------------|---------|-------------|
| [current folder]                   |   |   |   | 2026-08-17 22:02:46 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 22:02:46 UTC | 4096  | Allocated   | unknown | Allocated   |
| [parent folder]                    |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 07:30:00 UTC | 2026-08-18 07:30:00 UTC | 4096  | Allocated   | unknown | Allocated   |
| B412_NOT_SERVICEABLE.pdf           |   |   |   | 2026-08-17 22:02:46 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 22:02:46 UTC | 2361  | Unallocated | unknown | Unallocated |
| Component_Certificate_Original.pdf |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 2268  | Unallocated | unknown | Unallocated |
| Diagnostic_Export_15Aug2026.csv    |   |   |   | 2026-08-15 19:02:14 UTC | 0000-00-00 00:00:00 | 2026-08-15 00:00:00 UTC | 2026-08-15 19:02:14 UTC | 1438  | Unallocated | unknown | Unallocated |
| Gearbox_Inspection_Notes.txt       |   |   |   | 2026-08-17 22:02:46 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 22:02:46 UTC | 207   | Unallocated | unknown | Unallocated |
| Maintenance_History_Backup.db      |   |   |   | 2026-08-17 22:02:46 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 22:02:46 UTC | 12288 | Unallocated | unknown | Unallocated |
| Supplier_Component_Query.eml       |   |   |   | 2026-08-17 19:21:00 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 19:21:00 UTC | 275   | Unallocated | unknown | Unallocated |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$orphanFiles (0)
      - \$recycle.Bin (2)
      - \$I50allot (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - XavierAir (3)
      - USB\_Backups (3)
      - XAL\_MAINT\_16GB (8)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)
          - Temporary\_Review (8)
          - Downloads (5)
          - Pictures (3)
          - Windows (4)
          - INF (4)
          - system32 (3)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Windows/INF

Table Thumbnail Summary

Save Table as CSV

| Name                | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(Meta) | Loc |
|---------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-------------|-----|
| [current folder]    |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096 | Allocated  | unknown | Allocated   | /im |
| [parent folder]     |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096 | Allocated  | unknown | Allocated   | /im |
| setupapi.dev.log    |   |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 391  | Allocated  | unknown | Allocated   | /im |
| USB_CONNECTIONS.csv |   |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 431  | Allocated  | unknown | Allocated   | /im |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Go to Page Script: Latin - Basic

2026-08-15T18:14:08Z USBSTOR:Disk&Ven\_Kingston&Prod\_DataTraveler 3.0 Serial=60A44C41A21F8F10 Users=kwame.adu

2026-07-22T10:10:44Z USBSTOR:Disk&Ven\_Samsung&Prod\_Ultra USB 3.0 Serial=4C33000112092111830Z Users=engineering.admin

2026-08-16T12:05:17Z USBSTOR:Disk&Ven\_Samsung&Prod\_Android MTP Serial=K5BW80XAL01 Users=kwame.adu

CREATED BY HIVE CONSULT, NANA SEI ANYEMEDU / PROPERTY OF HIVE CONSULT

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$orphanFiles (0)
      - \$recycle.Bin (2)
      - \$I50allot (2)
      - Case\_Evidence (18)
      - ProgramData (3)
      - XavierAir (3)
      - USB\_Backups (3)
      - XAL\_MAINT\_16GB (8)
      - Users (3)
        - kwame.adu (7)
          - AppData (4)
          - Desktop (2)
          - Documents (9)
          - Temporary\_Review (8)
          - Downloads (5)
          - Pictures (3)
          - Windows (4)
          - INF (4)
          - system32 (3)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol\_vol2/Windows/INF

Table Thumbnail Summary

Save Table as CSV

| Name                | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(Meta) | Loc |
|---------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-------------|-----|
| [current folder]    |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096 | Allocated  | unknown | Allocated   | /im |
| [parent folder]     |   |   |   | 2026-08-18 07:30:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 07:30:00 UTC | 4096 | Allocated  | unknown | Allocated   | /im |
| setupapi.dev.log    |   |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 391  | Allocated  | unknown | Allocated   | /im |
| USB_CONNECTIONS.csv |   |   |   | 2026-08-17 21:53:10 UTC | 0000-00-00 00:00:00 | 2026-08-17 00:00:00 UTC | 2026-08-17 21:53:10 UTC | 431  | Allocated  | unknown | Allocated   | /im |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text

volume\_label.vendor.product.serial.drive\_letter.first\_connected\_utc.user

XAL\_MAINT\_16GB Kingston DataTraveler 3.0 60A44C41A21F8F10 E: 2026-08-15T18:14:08Z 2026-08-17T22:03:12Z kwame.adu

ENGINEERING\_BACKUP: SanDisk Ultra USB 3.0 4C33000112092111830Z F: 2026-07-22T10:10:44Z 2026-08-10T16:33:02Z engineering.admin

SAMSUNG\_A24 Samsung Android MTP K5BW80XAL01 2026-08-16T12:05:17Z 2026-08-16T12:22:48Z kwame.adu

-----METADATA-----

Content-Encoding: windows-1252

Content-Type: text/plain; charset=windows-1252

X-TIKA:Parsed-By: org.apache.tika.parser.DefaultParser

X-TIKA:Parsed-By-Full-Set: org.apache.tika.parser.DefaultParser

X-TIKA:detectedEncoding: windows-1252

X-TIKA:encodingDetector: UniversalEncodingDetector

**What the USB Evidence Revealed: Analysis of the recovered USB device XAL\_MAINT\_16GB revealed copies of maintenance documentation, component records, and supporting evidence relating to helicopter XAL-412. The USB contained the Original Maintenance Report which is B412\_Maintenance\_Status\_15Aug2026\_FINAL.docx, This document appears to be the original maintenance status report that existed before later modifications to the maintenance records.**

|                                        |
|----------------------------------------|
| <b>Component Documentation</b>         |
| Component_Certificate_XAL-412-7781.pdf |
| Installed_Component_Serial.jpg         |
| Installed_Component_Serial_Closeup.png |
| Supplier_Invoice_XAL-412-7781.pdf      |

**These files provided evidence of the installed gearbox oil-pressure sensor and allowed comparison of the installed serial number against the component certificate and supplier invoice, ultimately revealing the serial number mismatch.**

**Within C:\Users\kwame.adu\Documents\Temporary\_Review, several deleted files were recovered:**

|                                                |
|------------------------------------------------|
| <b>Deleted Files Recovered from the Laptop</b> |
| B412_NOT_SERVICEABLE.pdf                       |
| Component_Certificate_Original.pdf             |
| Diagnostic_Export_15Aug2026.csv                |
| Gearbox_Inspection_Notes.txt                   |
| Maintenance_History_Backup.db                  |
| Supplier_Component_Query.eml                   |

**These deleted artifacts indicate that earlier maintenance records, diagnostic exports, inspection notes, and supporting documentation existed on the system but were subsequently removed.**

**When Was the USB Connected? The USB connection artifacts provide direct evidence of device usage.**

**USB\_CONNECTIONS.csv**

|                 |                      |
|-----------------|----------------------|
| Volume Label    | XAL_MAINT_16GB       |
| Vendor          | Kingston             |
| Product         | DataTraveler 3.0     |
| Serial          | 60A44C421A21F8F10    |
| User            | kwame.adu            |
| First Connected | 2026-08-15T18:14:08Z |
| Last Connected  | 2026-08-17T22:03:12Z |
| Drive Letter    | K:                   |

**The SetupAPI and USB connection records corroborate that the Kingston DataTraveler 3.0 USB device (XAL\_MAINT\_16GB) was connected by the kwame.adu account.**

6. Using the telemetry, HUMS and black-box/FDR data, reconstruct the sequence in which gearbox temperature, oil pressure and vibration changed during the final stage of the flight. Which abnormal condition appeared first?

**Ans: Correlation of the HUMS, Flight Telemetry, and FDR datasets indicates that the first abnormal condition was a rise in gearbox temperature, which exceeded 100°C at 08:31:59 UTC. Approximately two minutes later, rotor vibration exceeded 1.0 g at 08:34:09 UTC, indicating increasing mechanical instability. Finally, oil pressure fell below 30 psi at 08:34:34 UTC, confirming a critical lubrication deficiency. The sequence of events suggests progressive gearbox degradation characterized by overheating, increasing vibration, and loss of lubrication prior to the helicopter's rapid descent and eventual crash.**

### Step 1: First Abnormal Condition

#### a. Evidence from FDR (Artifact: B412\_FDR\_Parsed\_18AUG2026.csv)

At 08:31:59 UTC, gearbox temperature exceeded 100°C for the first time.

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:31:59 | 100.10            | 35.16              | 0.79          |

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:31:59 | 100.10            | 35.16              | 0.79          |

#### b. Evidence from Flight Telemetry (Artifact: B412\_Flight\_Telemetry\_18AUG2026.csv)

This independently confirms the same event.

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:31:59 | 100.10            | 35.16              | 0.79          |

|                     |        |       |      |
|---------------------|--------|-------|------|
| 2026-08-18 08:31:59 | 100.08 | 35.13 | 0.81 |
|---------------------|--------|-------|------|

|                                 |                      |          |          |        |      |       |   |       |       |        |        |       |       |        |            |        |  |  |  |  |  |  |  |  |  |
|---------------------------------|----------------------|----------|----------|--------|------|-------|---|-------|-------|--------|--------|-------|-------|--------|------------|--------|--|--|--|--|--|--|--|--|--|
| B412_Flight_Telemetry_18AUG2026 |                      |          |          |        |      |       |   |       |       |        |        |       |       |        |            |        |  |  |  |  |  |  |  |  |  |
| 346                             | 2026-08-18T08:30:54Z | 5.20093  | -1.57114 | 5062.3 | 97   | 245.7 | 0 | 73.64 | 95.63 | 99.65  | 97.35  | 37.08 | 0.696 | 2977.2 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 347                             | 2026-08-18T08:30:59Z | 5.201136 | -1.57513 | 5058.9 | 96.8 | 245.8 | 0 | 73.69 | 95.62 | 99.66  | 97.01  | 36.82 | 0.698 | 2971.9 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 348                             | 2026-08-18T08:31:04Z | 5.201409 | -1.57911 | 5055.3 | 96.7 | 245.8 | 0 | 73.73 | 95.6  | 99.68  | 97.8   | 36.65 | 0.739 | 2971.7 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 349                             | 2026-08-18T08:31:09Z | 5.201749 | -1.58309 | 5051.6 | 96.6 | 245.9 | 0 | 73.77 | 95.59 | 99.7   | 97.51  | 36.59 | 0.739 | 2961   | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 350                             | 2026-08-18T08:31:14Z | 5.202154 | -1.58706 | 5047.9 | 96.5 | 246.9 | 0 | 73.8  | 95.57 | 99.71  | 97.82  | 36.1  | 0.721 | 2973.4 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 351                             | 2026-08-18T08:31:19Z | 5.202626 | -1.59103 | 5043.9 | 96.3 | 245.9 | 0 | 73.84 | 95.56 | 99.73  | 98.24  | 36    | 0.736 | 2965.1 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 352                             | 2026-08-18T08:31:24Z | 5.203163 | -1.59499 | 5039.9 | 96.2 | 246   | 0 | 73.86 | 95.55 | 99.76  | 98.42  | 36.23 | 0.753 | 2970.4 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 353                             | 2026-08-18T08:31:29Z | 5.203765 | -1.59894 | 5035.9 | 96.1 | 246   | 0 | 73.89 | 95.54 | 99.78  | 98.75  | 36.1  | 0.747 | 2962.6 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 354                             | 2026-08-18T08:31:34Z | 5.204431 | -1.60289 | 5031.8 | 96   | 246.1 | 0 | 73.92 | 95.53 | 99.8   | 98.74  | 36.02 | 0.762 | 2960.7 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 355                             | 2026-08-18T08:31:39Z | 5.205161 | -1.60683 | 5027.6 | 95.8 | 246.1 | 0 | 73.94 | 95.52 | 99.83  | 99     | 36.06 | 0.769 | 2977.4 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 356                             | 2026-08-18T08:31:44Z | 5.205955 | -1.61077 | 5023.4 | 95.7 | 246.2 | 0 | 73.96 | 95.52 | 99.85  | 98.9   | 35.96 | 0.792 | 2974.3 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 357                             | 2026-08-18T08:31:49Z | 5.206811 | -1.6147  | 5019.1 | 95.6 | 246.2 | 0 | 73.97 | 95.51 | 99.88  | 99.11  | 35.38 | 0.797 | 2971.6 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 358                             | 2026-08-18T08:31:54Z | 5.207729 | -1.61863 | 5014.8 | 95.4 | 246.3 | 0 | 73.98 | 95.51 | 99.91  | 99.78  | 35.39 | 0.809 | 2973.6 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 359                             | 2026-08-18T08:31:59Z | 5.208709 | -1.62255 | 5010.5 | 95.3 | 246.3 | 0 | 73.99 | 95.5  | 99.93  | 100.08 | 35.13 | 0.808 | 2959.5 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 360                             | 2026-08-18T08:32:04Z | 5.20975  | -1.62646 | 5006.1 | 95.2 | 246.4 | 0 | 74    | 95.5  | 99.96  | 100.32 | 34.46 | 0.818 | 2952.1 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 361                             | 2026-08-18T08:32:09Z | 5.210852 | -1.63037 | 5001.7 | 95.1 | 246.4 | 0 | 74    | 95.5  | 99.99  | 100    | 34.85 | 0.831 | 2959.8 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 362                             | 2026-08-18T08:32:14Z | 5.212012 | -1.63428 | 4997.4 | 94.9 | 246.5 | 0 | 74    | 95.5  | 100.02 | 100.8  | 34.55 | 0.81  | 2975.6 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 363                             | 2026-08-18T08:32:19Z | 5.213232 | -1.63817 | 4993   | 94.8 | 246.6 | 0 | 74    | 95.5  | 100.04 | 100.57 | 34.67 | 0.807 | 2967.4 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 364                             | 2026-08-18T08:32:24Z | 5.21451  | -1.64207 | 4988.7 | 94.7 | 246.6 | 0 | 73.99 | 95.5  | 100.07 | 100.91 | 33.9  | 0.838 | 2963.3 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 365                             | 2026-08-18T08:32:29Z | 5.215845 | -1.64595 | 4984.3 | 94.5 | 246.7 | 0 | 73.98 | 95.51 | 100.1  | 101.32 | 33.55 | 0.831 | 2963.7 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 366                             | 2026-08-18T08:32:34Z | 5.217237 | -1.64984 | 4980   | 94.4 | 246.7 | 0 | 73.97 | 95.51 | 100.13 | 101.38 | 33.74 | 0.861 | 2969.8 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 367                             | 2026-08-18T08:32:39Z | 5.218685 | -1.65371 | 4975.8 | 94.3 | 246.8 | 0 | 73.95 | 95.52 | 100.15 | 101.43 | 33.65 | 0.848 | 2962   | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 368                             | 2026-08-18T08:32:44Z | 5.220188 | -1.65758 | 4971.5 | 94.1 | 246.8 | 0 | 73.93 | 95.52 | 100.18 | 101.91 | 33.16 | 0.86  | 2969.2 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 369                             | 2026-08-18T08:32:49Z | 5.221745 | -1.66145 | 4967.4 | 94   | 246.9 | 0 | 73.91 | 95.53 | 100.2  | 102.13 | 32.97 | 0.884 | 2963.5 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 370                             | 2026-08-18T08:32:54Z | 5.223355 | -1.66531 | 4963.3 | 93.9 | 247   | 0 | 73.89 | 95.54 | 100.23 | 102.02 | 32.73 | 0.881 | 2966.8 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 371                             | 2026-08-18T08:32:59Z | 5.225018 | -1.66916 | 4959.3 | 93.8 | 247   | 0 | 73.86 | 95.55 | 100.25 | 102.43 | 33.25 | 0.902 | 2957.5 | ROTOR, VII | NORMAL |  |  |  |  |  |  |  |  |  |
| 372                             | 2026-08-18T08:33:04Z | 5.226733 | -1.67301 | 4955.3 | 93.6 | 247.1 | 0 | 73.83 | 95.56 | 100.27 | 102.54 | 32.91 | 0.896 | 2970.1 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 373                             | 2026-08-18T08:33:09Z | 5.228498 | -1.67686 | 4951.4 | 93.5 | 247.1 | 0 | 73.8  | 95.58 | 100.29 | 102.69 | 32.76 | 0.907 | 2965.5 | ROTOR, VII | NORMAL |  |  |  |  |  |  |  |  |  |
| 374                             | 2026-08-18T08:33:14Z | 5.230313 | -1.6807  | 4947.7 | 93.4 | 247.2 | 0 | 73.76 | 95.59 | 100.31 | 103.02 | 32.38 | 0.893 | 2961.1 | NONE       | NORMAL |  |  |  |  |  |  |  |  |  |
| 375                             | 2026-08-18T08:33:19Z | 5.232176 | -1.68453 | 4944   | 93.3 | 247.3 | 0 | 73.72 | 95.6  | 100.33 | 103.15 | 32.65 | 0.924 | 2957.2 | ROTOR, VII | NORMAL |  |  |  |  |  |  |  |  |  |
| 376                             | 2026-08-18T08:33:24Z | 5.234088 | -1.68836 | 4940.4 | 93.1 | 247.3 | 0 | 73.68 | 95.62 | 100.34 | 103.75 | 31.75 | 0.928 | 2968.8 | ROTOR, VII | NORMAL |  |  |  |  |  |  |  |  |  |
| 377                             | 2026-08-18T08:33:29Z | 5.236046 | -1.69218 | 4937   | 93   | 247.4 | 0 | 73.64 | 95.63 | 100.35 | 103.32 | 31.83 | 0.933 | 2966.1 | ROTOR, VII | NORMAL |  |  |  |  |  |  |  |  |  |
| 378                             | 2026-08-18T08:33:34Z | 5.23805  | -1.696   | 4933.6 | 92.9 | 247.5 | 0 | 73.59 | 95.65 | 100.37 | 103.6  | 31.31 | 0.941 | 2969.1 | ROTOR, VII | NORMAL |  |  |  |  |  |  |  |  |  |

c. Evidence from HUMS (Artifact: HUMS\_18AUG2026.csv)

HUMS samples every 10 seconds, so it records the same condition a few seconds later.

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:32:04 | 100.32            | 34.46              | 0.82          |

|     | A                    | B | C     | D      | E     | F                                                 | G | H | I | J | K |
|-----|----------------------|---|-------|--------|-------|---------------------------------------------------|---|---|---|---|---|
| 172 | 2026-08-18T08:30:34Z |   | 0.665 | 95.92  | 37.63 | 73.45 NONE                                        |   |   |   |   |   |
| 173 | 2026-08-18T08:30:44Z |   | 0.686 | 96.66  | 37.49 | 73.55 NONE                                        |   |   |   |   |   |
| 174 | 2026-08-18T08:30:54Z |   | 0.696 | 97.35  | 37.08 | 73.64 NONE                                        |   |   |   |   |   |
| 175 | 2026-08-18T08:31:04Z |   | 0.739 | 97.8   | 36.65 | 73.73 NONE                                        |   |   |   |   |   |
| 176 | 2026-08-18T08:31:14Z |   | 0.721 | 97.82  | 36.1  | 73.8 NONE                                         |   |   |   |   |   |
| 177 | 2026-08-18T08:31:24Z |   | 0.753 | 98.42  | 36.23 | 73.86 NONE                                        |   |   |   |   |   |
| 178 | 2026-08-18T08:31:34Z |   | 0.762 | 98.74  | 36.02 | 73.92 NONE                                        |   |   |   |   |   |
| 179 | 2026-08-18T08:31:44Z |   | 0.792 | 98.9   | 35.96 | 73.96 NONE                                        |   |   |   |   |   |
| 180 | 2026-08-18T08:31:54Z |   | 0.809 | 99.78  | 35.39 | 73.98 NONE                                        |   |   |   |   |   |
| 181 | 2026-08-18T08:32:04Z |   | 0.818 | 100.32 | 34.46 | 74 NONE                                           |   |   |   |   |   |
| 182 | 2026-08-18T08:32:14Z |   | 0.81  | 100.8  | 34.55 | 74 NONE                                           |   |   |   |   |   |
| 183 | 2026-08-18T08:32:24Z |   | 0.838 | 100.91 | 33.9  | 73.99 NONE                                        |   |   |   |   |   |
| 184 | 2026-08-18T08:32:34Z |   | 0.861 | 101.38 | 33.74 | 73.97 NONE                                        |   |   |   |   |   |
| 185 | 2026-08-18T08:32:44Z |   | 0.86  | 101.91 | 33.16 | 73.93 NONE                                        |   |   |   |   |   |
| 186 | 2026-08-18T08:32:54Z |   | 0.881 | 102.02 | 32.73 | 73.89 NONE                                        |   |   |   |   |   |
| 187 | 2026-08-18T08:33:04Z |   | 0.896 | 102.54 | 32.91 | 73.83 NONE                                        |   |   |   |   |   |
| 188 | 2026-08-18T08:33:14Z |   | 0.893 | 103.02 | 32.38 | 73.76 NONE                                        |   |   |   |   |   |
| 189 | 2026-08-18T08:33:24Z |   | 0.928 | 103.75 | 31.75 | 73.68 ROTOR_VIB_ALERT                             |   |   |   |   |   |
| 190 | 2026-08-18T08:33:34Z |   | 0.941 | 103.6  | 31.31 | 73.59 ROTOR_VIB_ALERT                             |   |   |   |   |   |
| 191 | 2026-08-18T08:33:44Z |   | 0.957 | 104.03 | 31.9  | 73.49 ROTOR_VIB_ALERT                             |   |   |   |   |   |
| 192 | 2026-08-18T08:33:54Z |   | 0.966 | 105.02 | 31.16 | 73.38 GBX_TEMP_HIGH;ROTOR_VIB_ALERT               |   |   |   |   |   |
| 193 | 2026-08-18T08:34:04Z |   | 0.975 | 104.77 | 31.15 | 73.26 ROTOR_VIB_ALERT                             |   |   |   |   |   |
| 194 | 2026-08-18T08:34:14Z |   | 0.989 | 105.88 | 30.17 | 73.13 GBX_TEMP_HIGH;ROTOR_VIB_ALERT               |   |   |   |   |   |
| 195 | 2026-08-18T08:34:24Z |   | 0.992 | 105.76 | 30.5  | 73 GBX_TEMP_HIGH;ROTOR_VIB_ALERT                  |   |   |   |   |   |
| 196 | 2026-08-18T08:34:34Z |   | 1.035 | 106.45 | 29.82 | 72.86 GBX_TEMP_HIGH;OIL_PRESS_LOW;ROTOR_VIB_ALERT |   |   |   |   |   |
| 197 | 2026-08-18T08:34:44Z |   | 1.046 | 106.88 | 29.66 | 72.72 GBX_TEMP_HIGH;OIL_PRESS_LOW;ROTOR_VIB_ALERT |   |   |   |   |   |
| 198 | 2026-08-18T08:34:54Z |   | 1.071 | 107.48 | 28.66 | 72.57 GBX_TEMP_HIGH;OIL_PRESS_LOW;ROTOR_VIB_ALERT |   |   |   |   |   |
| 199 | 2026-08-18T08:35:04Z |   | 1.061 | 107.47 | 28.69 | 72.42 GBX_TEMP_HIGH;OIL_PRESS_LOW;ROTOR_VIB_ALERT |   |   |   |   |   |

**Finding: The first abnormal condition was excessive gearbox temperature. This was first observed in the FDR and Flight Telemetry datasets at 08:31:59 UTC, when gearbox temperature exceeded 100°C.**

## Step 2: Vibration Becomes Abnormal

### a. Evidence from FDR (Artifact: B412\_FDR\_Parsed\_18AUG2026.csv)

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:34:09 | 105.47            | 30.63              | 1.003         |

AutoSave On B412\_FDR\_Parsed\_18AUG2026

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard Font Alignment Number Styles Cells Editing

Comments Share

2026-08-18T08:34:09Z

| A    | B                    | C        | D        | E        | F        | G           | H            | I           | J           | K           | L           | M           | N            | O                             | P | Q | R | S |
|------|----------------------|----------|----------|----------|----------|-------------|--------------|-------------|-------------|-------------|-------------|-------------|--------------|-------------------------------|---|---|---|---|
| 1908 | 2026-08-18T08:34:00Z | 5.248721 | -1.71502 | 4918.978 | 92.27051 | 247.8183009 | 0.007454924  | 73.34359775 | 95.74061467 | 100.4099894 | 104.7426727 | 30.43148094 | 0.963004549  | ROTOR_VIB_ALERT               |   |   |   |   |
| 1909 | 2026-08-18T08:34:01Z | 5.248721 | -1.71502 | 4919.017 | 92.30222 | 247.8109611 | -0.011777235 | 73.32404998 | 95.76370788 | 100.3764244 | 104.7525213 | 30.38510045 | 0.941930759  | ROTOR_VIB_ALERT               |   |   |   |   |
| 1910 | 2026-08-18T08:34:02Z | 5.248721 | -1.71502 | 4919.013 | 92.27955 | 247.7713724 | -0.01091813  | 73.3361083  | 95.77136742 | 100.4229599 | 104.7681636 | 30.41575399 | 0.974720099  | ROTOR_VIB_ALERT               |   |   |   |   |
| 1911 | 2026-08-18T08:34:03Z | 5.248721 | -1.71502 | 4918.993 | 92.28379 | 247.8003711 | 0.019864151  | 73.32358196 | 95.76906095 | 100.3773703 | 104.7675534 | 30.4253971  | 0.94305179   | ROTOR_VIB_ALERT               |   |   |   |   |
| 1912 | 2026-08-18T08:34:04Z | 5.250978 | -1.71881 | 4916.472 | 92.22571 | 247.9256635 | -0.02792665  | 73.28589727 | 95.7441446  | 100.377594  | 104.78767   | 31.17708356 | 0.948034767  | ROTOR_VIB_ALERT               |   |   |   |   |
| 1913 | 2026-08-18T08:34:05Z | 5.250978 | -1.71881 | 4916.495 | 92.18614 | 247.9208089 | -0.01177479  | 73.25524411 | 95.76939614 | 100.405312  | 104.7166688 | 31.1673095  | 0.90703015   | ROTOR_VIB_ALERT               |   |   |   |   |
| 1914 | 2026-08-18T08:34:06Z | 5.250978 | -1.71881 | 4916.496 | 92.18006 | 247.8934454 | -0.015051609 | 73.2363547  | 95.7517177  | 100.4243462 | 104.772673  | 31.1730511  | 0.97592381   | ROTOR_VIB_ALERT               |   |   |   |   |
| 1915 | 2026-08-18T08:34:07Z | 5.250978 | -1.71881 | 4916.524 | 92.18731 | 247.883241  | -0.006519058 | 73.26155159 | 95.75540863 | 100.3957929 | 104.7948903 | 31.15810841 | 0.966839461  | ROTOR_VIB_ALERT               |   |   |   |   |
| 1916 | 2026-08-18T08:34:08Z | 5.250978 | -1.71881 | 4916.492 | 92.17813 | 247.9002627 | -0.024629533 | 73.24756478 | 95.74410261 | 100.4278588 | 104.7745115 | 31.1333586  | 0.970777287  | ROTOR_VIB_ALERT               |   |   |   |   |
| 1917 | 2026-08-18T08:34:09Z | 5.252373 | -1.72259 | 4914.188 | 92.10978 | 247.9111697 | 0.01986159   | 73.2181117  | 95.76465489 | 100.3792734 | 105.4715041 | 30.62708988 | 1.003091996  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1918 | 2026-08-18T08:34:10Z | 5.252373 | -1.72259 | 4914.192 | 92.10308 | 247.9236684 | -0.001180688 | 73.19705527 | 95.81753132 | 100.4286312 | 105.4863445 | 30.5736315  | 0.962549646  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1919 | 2026-08-18T08:34:11Z | 5.252373 | -1.72259 | 4914.214 | 92.12342 | 247.8829722 | -0.010768223 | 73.18991118 | 95.76895161 | 100.3996309 | 105.4629902 | 30.57053457 | 1.004055379  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1920 | 2026-08-18T08:34:12Z | 5.252373 | -1.72259 | 4914.219 | 92.06793 | 247.8934464 | -0.014302749 | 73.21649019 | 95.80934565 | 100.4247047 | 105.4762624 | 30.61034965 | 0.993495923  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1921 | 2026-08-18T08:34:13Z | 5.252373 | -1.72259 | 4914.19  | 92.12444 | 247.8884837 | -0.013035666 | 73.20746001 | 95.81905188 | 100.396635  | 105.4696107 | 30.57599157 | 0.9383031446 | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1922 | 2026-08-18T08:34:14Z | 5.255605 | -1.72637 | 4912.01  | 91.97231 | 247.9741952 | 0.003518088  | 73.14635047 | 95.83374852 | 100.3969541 | 105.8830075 | 30.16360871 | 1.00849694   | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1923 | 2026-08-18T08:34:15Z | 5.255605 | -1.72637 | 4912.003 | 91.99    | 247.9786804 | 0.005745745  | 73.10423111 | 95.79705596 | 100.3990904 | 105.8869199 | 30.15645635 | 0.974368724  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1924 | 2026-08-18T08:34:16Z | 5.255605 | -1.72637 | 4912.006 | 92.0106  | 247.9811506 | -0.001838051 | 73.12630323 | 95.82946824 | 100.3784315 | 105.8901197 | 30.15287634 | 0.991115971  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1925 | 2026-08-18T08:34:17Z | 5.255605 | -1.72637 | 4912.015 | 92.01523 | 248.0117071 | -0.01671711  | 73.12471086 | 95.80930272 | 100.4057766 | 105.9073027 | 30.18196838 | 1.013203663  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1926 | 2026-08-18T08:34:18Z | 5.255605 | -1.72637 | 4911.995 | 92.00436 | 248.0201529 | -0.00735543  | 73.14946414 | 95.82541635 | 100.428409  | 105.8798911 | 30.1567297  | 1.009807837  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1927 | 2026-08-18T08:34:19Z | 5.257973 | -1.73014 | 4909.001 | 91.8812  | 248.0717785 | -0.01066686  | 73.04227865 | 95.8551787  | 100.403096  | 106.0149124 | 30.33583475 | 0.991031728  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1928 | 2026-08-18T08:34:20Z | 5.257973 | -1.73014 | 4909.996 | 91.9148  | 248.0880247 | -0.022510696 | 73.07675422 | 95.83260331 | 100.4185064 | 106.0205241 | 30.3056149  | 1.004017747  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1929 | 2026-08-18T08:34:21Z | 5.257973 | -1.73014 | 4910.005 | 91.87757 | 248.0890471 | -0.01833763  | 73.0585867  | 95.81601805 | 100.361119  | 106.0464359 | 30.3108283  | 0.987777982  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1930 | 2026-08-18T08:34:22Z | 5.257973 | -1.73014 | 4909.979 | 91.81599 | 248.0860747 | 0.01585462   | 73.06027749 | 95.85212727 | 100.3620013 | 106.031103  | 30.33487523 | 1.017588696  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1931 | 2026-08-18T08:34:23Z | 5.257973 | -1.73014 | 4909.978 | 91.96872 | 248.1286477 | -0.01061759  | 73.05683361 | 95.81862353 | 100.3625672 | 106.0256564 | 30.30928623 | 1.003036796  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1932 | 2026-08-18T08:34:24Z | 5.260376 | -1.73391 | 4908.103 | 91.76856 | 248.2207565 | -0.002715969 | 72.97814674 | 95.86554127 | 100.3756143 | 105.760613  | 30.50739106 | 0.978364589  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1933 | 2026-08-18T08:34:25Z | 5.260376 | -1.73391 | 4908.171 | 91.82267 | 248.1756427 | -0.01315971  | 73.01539285 | 95.83902612 | 100.3784538 | 105.7510222 | 30.49882143 | 1.008979977  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1934 | 2026-08-18T08:34:26Z | 5.260376 | -1.73391 | 4908.186 | 91.79238 | 248.1727719 | 0.021919366  | 73.0130001  | 95.88494132 | 100.3995715 | 105.7677799 | 30.51374687 | 0.982329585  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1935 | 2026-08-18T08:34:27Z | 5.260376 | -1.73391 | 4908.201 | 91.78334 | 248.2234009 | -0.02004072  | 72.9851666  | 95.85852054 | 100.3968375 | 105.7388882 | 30.51668185 | 0.980219613  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1936 | 2026-08-18T08:34:28Z | 5.260376 | -1.73391 | 4908.194 | 91.81577 | 248.2084872 | 0.019441132  | 72.98339475 | 95.88478916 | 100.3638127 | 105.7518373 | 30.47511109 | 0.991131959  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1937 | 2026-08-18T08:34:29Z | 5.262812 | -1.73768 | 4906.597 | 91.70445 | 248.154254  | -0.00408211  | 72.94400929 | 95.87308579 | 100.3607871 | 106.1152397 | 30.36408386 | 1.004056238  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1938 | 2026-08-18T08:34:30Z | 5.262812 | -1.73768 | 4906.619 | 91.67557 | 248.2031879 | 0.016471546  | 72.91782621 | 95.8500221  | 100.3893668 | 106.1424741 | 30.36189125 | 1.032431528  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1939 | 2026-08-18T08:34:31Z | 5.262812 | -1.73768 | 4906.571 | 91.69437 | 248.2090025 | -0.006271597 | 72.93886184 | 95.88861804 | 100.355666  | 106.1453635 | 30.33267828 | 1.018726962  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |
| 1940 | 2026-08-18T08:34:32Z | 5.262812 | -1.73768 | 4906.593 | 91.71418 | 248.2011949 | -0.017480258 | 72.96819473 | 95.87434024 | 100.3710532 | 106.1565912 | 30.3607465  | 1.025424312  | GBX_TEMP_HIGH,ROTOR_VIB_ALERT |   |   |   |   |

B412\_FDR\_Parsed\_18AUG2026

Ready Accessibility: Unavailable

**b. Evidence from Telemetry (Artifact: B412\_Flight\_Telemetry\_18AUG2026.csv)**

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:34:09 | 105.49            | 30.60              | 1.007         |

AutoSave On

B412\_Flight\_Telemetry\_18AUG2026

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelp

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

ClipboardFontAlignmentNumberStylesCellsEditing

CommentsShare

### Step 3: Oil Pressure Reaches Critical Level

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:34:34 | 106.47            | 29.84              | 1.04          |

***b. Evidence from Telemetry (Artifact: B412\_Flight\_Telemetry\_18AUG2026.csv)***

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:34:34 | 106.45            | 29.82              | 1.04          |

AutoSave ON B412\_Flight\_Telemetry\_18AUG2026

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard Font Alignment Number Styles Cells Editing

Comments Share

|     | A                    | B        | C        | D      | E    | F     | G     | H     | I     | J      | K | L      | M     | N     | O      | P                | Q | R | S | T | U |
|-----|----------------------|----------|----------|--------|------|-------|-------|-------|-------|--------|---|--------|-------|-------|--------|------------------|---|---|---|---|---|
| 382 | 2026-08-18T08:33:54Z | 5.246504 | -1.71122 | 4921.6 | 92.4 | 247.7 | 0     | 73.38 | 95.73 | 100.4  |   | 105.02 | 31.16 | 0.966 | 2968.1 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 383 | 2026-08-18T08:33:58Z | 5.248721 | -1.71502 | 4919   | 92.3 | 247.8 | 0     | 73.32 | 95.75 | 100.4  |   | 104.74 | 30.41 | 0.963 | 2959.1 | ROTOR_VIB_NORMAL |   |   |   |   |   |
| 384 | 2026-08-18T08:34:04Z | 5.250978 | -1.71881 | 4916.5 | 92.2 | 247.9 | 0     | 73.26 | 95.77 | 100.4  |   | 104.77 | 31.15 | 0.975 | 2961.8 | ROTOR_VIB_NORMAL |   |   |   |   |   |
| 385 | 2026-08-18T08:34:09Z | 5.253273 | -1.72259 | 4914.2 | 92.1 | 247.9 | 0     | 73.2  | 95.79 | 100.4  |   | 105.49 | 30.6  | 1.007 | 2961.4 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 386 | 2026-08-18T08:34:14Z | 5.255605 | -1.72637 | 4912   | 92   | 248   | 0     | 73.13 | 95.81 | 100.4  |   | 105.88 | 30.17 | 0.989 | 2973   | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 387 | 2026-08-18T08:34:18Z | 5.257973 | -1.73014 | 4910   | 91.9 | 248.1 | 0     | 73.07 | 95.83 | 100.39 |   | 106.04 | 30.32 | 1.014 | 2969.1 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 388 | 2026-08-18T08:34:24Z | 5.260376 | -1.73391 | 4908.2 | 91.8 | 248.2 | 0     | 73    | 95.86 | 100.38 |   | 105.76 | 30.5  | 0.992 | 2969.2 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 389 | 2026-08-18T08:34:29Z | 5.262812 | -1.73768 | 4906.6 | 91.7 | 248.2 | 0     | 72.93 | 95.88 | 100.37 |   | 105.17 | 30.35 | 1.027 | 2963.7 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 390 | 2026-08-18T08:34:34Z | 5.265281 | -1.74144 | 4905.1 | 91.6 | 248.3 | 0     | 72.86 | 95.9  | 100.36 |   | 106.45 | 29.82 | 1.035 | 2957.6 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 391 | 2026-08-18T08:34:39Z | 5.267782 | -1.74519 | 4903.8 | 91.5 | 248.4 | 0     | 72.79 | 95.93 | 100.35 |   | 106.4  | 29.22 | 1.052 | 2971.6 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 392 | 2026-08-18T08:34:44Z | 5.270312 | -1.74894 | 4902.7 | 91.4 | 248.4 | 0     | 72.72 | 95.95 | 100.34 |   | 106.88 | 29.68 | 1.046 | 2962.9 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 393 | 2026-08-18T08:34:49Z | 5.272871 | -1.75269 | 4901.8 | 91.3 | 248.5 | 0     | 72.64 | 95.97 | 100.32 |   | 106.61 | 28.81 | 1.066 | 2957.6 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 394 | 2026-08-18T08:34:54Z | 5.275457 | -1.75643 | 4901.1 | 91.2 | 248.6 | 0     | 72.57 | 96    | 100.3  |   | 107.48 | 28.66 | 1.071 | 2953.2 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 395 | 2026-08-18T08:34:59Z | 5.27807  | -1.76017 | 4900.5 | 91.1 | 248.7 | 0     | 72.49 | 96.02 | 100.28 |   | 107.53 | 28.96 | 1.081 | 2967.8 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 396 | 2026-08-18T08:35:04Z | 5.280707 | -1.7639  | 4900.2 | 91.1 | 248.7 | 0     | 72.42 | 96.05 | 100.26 |   | 107.47 | 28.69 | 1.061 | 2967.3 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 397 | 2026-08-18T08:35:09Z | 5.283369 | -1.76763 | 4900   | 91   | 248.8 | 0     | 72.34 | 96.07 | 100.24 |   | 108.04 | 28.8  | 1.071 | 2968.2 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 398 | 2026-08-18T08:35:14Z | 5.286053 | -1.77135 | 4900   | 90.9 | 248.9 | 0     | 72.26 | 96.09 | 100.22 |   | 108.22 | 28.41 | 1.062 | 2964   | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 399 | 2026-08-18T08:35:19Z | 5.288758 | -1.77507 | 4900.3 | 90.8 | 249   | 0     | 72.18 | 96.12 | 100.2  |   | 108.58 | 27.9  | 1.092 | 2964.4 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 400 | 2026-08-18T08:35:24Z | 5.291483 | -1.77879 | 4900.7 | 90.8 | 249   | 0     | 72.1  | 96.14 | 100.17 |   | 108.61 | 27.5  | 1.106 | 2969.8 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 401 | 2026-08-18T08:35:29Z | 5.294227 | -1.7825  | 4901.3 | 90.7 | 249.1 | 0     | 72.03 | 96.16 | 100.15 |   | 108.73 | 27.5  | 1.121 | 2963.5 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 402 | 2026-08-18T08:35:34Z | 5.296987 | -1.78621 | 4902.1 | 90.6 | 249.2 | 0     | 71.95 | 96.19 | 100.12 |   | 108.97 | 27.86 | 1.137 | 2967.1 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 403 | 2026-08-18T08:35:39Z | 5.299764 | -1.78991 | 4903   | 90.6 | 249.3 | 0     | 71.87 | 96.21 | 100.09 |   | 109.17 | 27.09 | 1.148 | 2951.9 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 404 | 2026-08-18T08:35:44Z | 5.302555 | -1.79361 | 4904.2 | 90.5 | 249.3 | 0     | 71.79 | 96.23 | 100.06 |   | 109.66 | 27.52 | 1.155 | 2963.6 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 405 | 2026-08-18T08:35:49Z | 5.30536  | -1.79731 | 4905.6 | 90.4 | 249.4 | 0     | 71.71 | 96.25 | 100.04 |   | 109.78 | 26.82 | 1.146 | 2957.8 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 406 | 2026-08-18T08:35:54Z | 5.308177 | -1.801   | 4907.1 | 90.4 | 249.5 | 0     | 71.63 | 96.27 | 100.01 |   | 110.15 | 26.53 | 1.17  | 2965.7 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 407 | 2026-08-18T08:35:59Z | 5.311004 | -1.80469 | 4908.8 | 90.3 | 249.6 | 0     | 71.56 | 96.29 | 99.98  |   | 110.07 | 26.39 | 1.174 | 2966.8 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 408 | 2026-08-18T08:36:04Z | 5.31384  | -1.80837 | 4910.7 | 90.3 | 249.6 | 0     | 71.48 | 96.31 | 99.95  |   | 110.23 | 26.93 | 1.189 | 2954.2 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 409 | 2026-08-18T08:36:09Z | 5.316685 | -1.81205 | 4912.7 | 90.3 | 249.7 | 0     | 71.41 | 96.33 | 99.93  |   | 110.67 | 26.73 | 1.18  | 2958   | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 410 | 2026-08-18T08:36:14Z | 5.319536 | -1.81573 | 4914.9 | 90.2 | 249.8 | 0     | 71.33 | 96.35 | 99.9   |   | 110.88 | 26.54 | 1.195 | 2951.9 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 411 | 2026-08-18T08:36:19Z | 5.322392 | -1.8194  | 4917.3 | 90.2 | 249.9 | 0     | 71.26 | 96.36 | 99.87  |   | 111.22 | 26.41 | 1.213 | 2951.4 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 412 | 2026-08-18T08:36:24Z | 5.325252 | -1.82307 | 4919.8 | 90.1 | 249.9 | 0     | 71.19 | 96.38 | 99.85  |   | 111.42 | 25.65 | 1.225 | 2956.2 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 413 | 2026-08-18T08:36:29Z | 5.328115 | -1.82674 | 4922.5 | 90.1 | 250   | 0     | 71.11 | 96.39 | 99.82  |   | 111.89 | 25.77 | 1.228 | 2949.4 | GBX_TEMP_NORMAL  |   |   |   |   |   |
| 414 | 2026-08-18T08:36:34Z | 5.330913 | -1.8304  | 4926.4 | 90.1 | 250.1 | -24.5 | 71.04 | 96.41 | 99.8   |   | 112.24 | 24.95 | 1.233 | 2949.3 | GBX_TEMP_NORMAL  |   |   |   |   |   |

Ready Accessibility: Unavailable

### c. Evidence from HUMS (Artifact: HUMS\_18AUG2026.csv)

| Timestamp (UTC)     | Gearbox Temp (°C) | Oil Pressure (psi) | Vibration (g) |
|---------------------|-------------------|--------------------|---------------|
| 2026-08-18 08:34:34 | 106.45            | 29.82              | 1.04          |

AutoSave ON HUMS\_18AUG2026

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard Font Alignment Number Styles Cells Editing

Comments Share

|     | A                    | B | C     | D      | E     | F     | G                                           | H | I | J | K |
|-----|----------------------|---|-------|--------|-------|-------|---------------------------------------------|---|---|---|---|
| 190 | 2026-08-18T08:33:34Z |   | 0.941 | 103.6  | 31.31 | 73.59 | ROTOR_VIB_ALERT                             |   |   |   |   |
| 191 | 2026-08-18T08:33:44Z |   | 0.957 | 104.03 | 31.9  | 73.49 | ROTOR_VIB_ALERT                             |   |   |   |   |
| 192 | 2026-08-18T08:33:54Z |   | 0.966 | 105.02 | 31.16 | 73.38 | GBX_TEMP_HIGH,ROTOR_VIB_ALERT               |   |   |   |   |
| 193 | 2026-08-18T08:34:04Z |   | 0.975 | 104.77 | 31.15 | 73.26 | ROTOR_VIB_ALERT                             |   |   |   |   |
| 194 | 2026-08-18T08:34:14Z |   | 0.989 | 105.88 | 30.17 | 73.13 | GBX_TEMP_HIGH,ROTOR_VIB_ALERT               |   |   |   |   |
| 195 | 2026-08-18T08:34:24Z |   | 0.992 | 105.76 | 30.5  | 73    | GBX_TEMP_HIGH,ROTOR_VIB_ALERT               |   |   |   |   |
| 196 | 2026-08-18T08:34:34Z |   | 1.035 | 106.45 | 29.82 | 72.86 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 197 | 2026-08-18T08:34:44Z |   | 1.046 | 106.88 | 29.66 | 72.72 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 198 | 2026-08-18T08:34:54Z |   | 1.071 | 107.48 | 28.66 | 72.57 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 199 | 2026-08-18T08:35:04Z |   | 1.061 | 107.47 | 28.69 | 72.42 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 200 | 2026-08-18T08:35:14Z |   | 1.082 | 108.22 | 28.41 | 72.26 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 201 | 2026-08-18T08:35:24Z |   | 1.106 | 108.61 | 27.5  | 72.1  | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 202 | 2026-08-18T08:35:34Z |   | 1.137 | 108.97 | 27.86 | 71.95 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 203 | 2026-08-18T08:35:44Z |   | 1.155 | 109.66 | 27.52 | 71.79 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 204 | 2026-08-18T08:35:54Z |   | 1.17  | 110.15 | 26.53 | 71.63 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 205 | 2026-08-18T08:36:04Z |   | 1.189 | 110.23 | 26.93 | 71.48 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 206 | 2026-08-18T08:36:14Z |   | 1.195 | 110.88 | 26.54 | 71.33 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 207 | 2026-08-18T08:36:24Z |   | 1.225 | 111.42 | 25.65 | 71.19 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 208 | 2026-08-18T08:36:34Z |   | 1.233 | 112.24 | 24.95 | 71.04 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 209 | 2026-08-18T08:36:44Z |   | 1.239 | 112.29 | 24.95 | 70.91 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 210 | 2026-08-18T08:36:54Z |   | 1.245 | 112.99 | 24.63 | 70.78 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 211 | 2026-08-18T08:37:04Z |   | 1.296 | 113.6  | 24.63 | 70.66 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 212 | 2026-08-18T08:37:14Z |   | 1.29  | 113.46 | 23.58 | 70.55 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 213 | 2026-08-18T08:37:24Z |   | 1.305 | 114.11 | 23.33 | 70.44 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 214 | 2026-08-18T08:37:34Z |   | 1.333 | 114.56 | 23.52 | 70.35 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 215 | 2026-08-18T08:37:44Z |   | 1.35  | 115.07 | 23.26 | 70.27 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 216 | 2026-08-18T08:37:54Z |   | 1.361 | 115.68 | 22.88 | 70.19 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |
| 217 | 2026-08-18T08:38:04Z |   | 1.389 | 116.35 | 21.82 | 70.13 | GBX_TEMP_HIGH,OIL_PRESS_LOW,ROTOR_VIB_ALERT |   |   |   |   |

Ready Accessibility: Unavailable

**Finding: Oil pressure dropped below the critical threshold of 30 psi at 08:34:34 UTC.**

#### **Final Reconstruction**

| <b>Order</b> | <b>Event</b>                       | <b>Time (UTC)</b> | <b>Source Artifact</b>       |
|--------------|------------------------------------|-------------------|------------------------------|
| 1            | Gearbox temperature exceeded 100°C | 08:31:59          | FDR & Flight Telemetry       |
| 2            | Rotor vibration exceeded 1.0 g     | 08:34:09          | FDR & Flight Telemetry       |
| 3            | Oil pressure dropped below 30 psi  | 08:34:34          | HUMS, FDR & Flight Telemetry |

7. What did Captain Daniel Mensah and the other crew member say during the final minutes of the Cockpit Voice Recorder, and how does their conversation correlate with the FDR, HUMS and telemetry data?

**Ans: The CVR shows the crew noticing a worsening gearbox problem during the final minutes of the flight. At 08:34:07 UTC, the crew reported that the gearbox temperature was increasing, which matches the rising temperature recorded in the FDR and Flight Telemetry data. At 08:35:32 UTC, they observed that oil pressure was dropping, consistent with the decreasing oil pressure recorded in the HUMS, FDR, and Telemetry datasets.**

**At 08:37:41 UTC, the crew reported noticeable vibration, corresponding with the increasing vibration values recorded in the HUMS and FDR data. By 08:38:18 UTC, they noted that the oil pressure was dropping rapidly, and at 08:39:01 UTC, Captain Mensah reported serious vibration, indicating that the gearbox condition had significantly deteriorated.**

**The crew's observations closely match the aircraft data, showing a sequence of rising gearbox temperature, falling oil pressure, and increasing vibration, which ultimately led to severe mechanical instability before the end of the recording at 08:40:22 UTC**

#### **Cockpit Voice Recorder (CVR) Analysis and Correlation with FDR, HUMS and Telemetry Data**

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img\_The Crashed Helicopter/img\_vol2/Case\_Evidence/03\_Black\_Box\_FDR\_CVR

Table Thumbnail Summary

Save Table as CSV

| Name                                    | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size    | Flags(Dir) | Known |
|-----------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|---------|------------|-------|
| [current folder]                        |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096    | Allocated  | unkno |
| [parent folder]                         |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096    | Allocated  | unkno |
| B412_CVR_18AUG2026.wav                  |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2968582 | Allocated  | unkno |
| B412_CVR_18AUG2026.wav.sha256.txt       |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 89      | Allocated  | unkno |
| B412_CVR_18AUG2026_COCKPIT_PLAYBACK.mp4 |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 1085141 | Allocated  | unkno |
| B412_CVR_Cockpit_Background.png         |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 453756  | Allocated  | unkno |
| B412_CVR_Transcript.csv                 |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 1282    | Allocated  | unkno |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset

Text Source: File Text

| time_utc | speaker | transcript                                                                                                     |
|----------|---------|----------------------------------------------------------------------------------------------------------------|
| 08:32:41 | PILOT   | I am getting that navigation indication again.                                                                 |
| 08:32:50 | PILOT   | Heading is stable. GPS is intermittently dropping.                                                             |
| 08:34:07 | CREW    | Gearbox temperature is climbing.                                                                               |
| 08:34:15 | PILOT   | I see it.                                                                                                      |
| 08:35:32 | CREW    | Oil pressure is moving down.                                                                                   |
| 08:35:40 | PILOT   | Watch it closely.                                                                                              |
| 08:36:12 | SYSTEM  | CAUTION TONE                                                                                                   |
| 08:36:16 | PILOT   | Gearbox oil pressure caution.                                                                                  |
| 08:36:24 | CREW    | Temperature still increasing.                                                                                  |
| 08:36:31 | PILOT   | Let us turn toward the nearest suitable landing area.                                                          |
| 08:37:04 | PILOT   | Accra Control, Xavier Four-One-Two, we are experiencing intermittent indications and requesting a lower level. |
| 08:37:41 | CREW    | Vibration.                                                                                                     |
| 08:37:44 | PILOT   | I feel it.                                                                                                     |
| 08:38:02 | SOUND   | INCREASED MECHANICAL NOISE                                                                                     |
| 08:38:10 | PILOT   | Torque is fluctuating.                                                                                         |
| 08:38:18 | CREW    | Oil pressure dropping fast.                                                                                    |
| 08:38:23 | PILOT   | Prepare for precautionary landing.                                                                             |
| 08:38:54 | SYSTEM  | WARNING TONE                                                                                                   |
| 08:39:01 | PILOT   | We have got a serious vibration now.                                                                           |
| 08:39:18 | SOUND   | LOUD MECHANICAL SOUND                                                                                          |

| Time (UTC) | CVR Event                                                              | Correlated Artifact                                                   | Interpretation                                                        |
|------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| 08:34:07   | Crew member:<br>"Gearbox temperature is climbing."                     | FDR / Telemetry:<br>Gearbox temperature increasing above normal trend | Crew observed the developing overheating condition.                   |
| 08:34:15   | Captain Mensah: "I see it."                                            | FDR / Telemetry                                                       | Confirms flight crew awareness of abnormal temperature indications.   |
| 08:35:32   | Crew member: "Oil pressure is moving down."                            | HUMS / FDR / Telemetry:<br>Oil pressure decreasing                    | Crew observed the lubrication system degradation.                     |
| 08:36:12   | CAUTION TONE                                                           | CVR + FDR Warning System                                              | Aircraft warning system activated due to abnormal parameters.         |
| 08:36:16   | Captain Mensah: "Gearbox oil pressure caution."                        | HUMS / FDR: Oil pressure approaching caution limits                   | Cockpit warning corresponds with recorded oil pressure deterioration. |
| 08:36:24   | Crew member: "Temperature still increasing."                           | FDR / Telemetry:<br>Continuing gearbox temperature rise               | Overheating condition persisted.                                      |
| 08:36:31   | Captain Mensah: "Let's turn toward the nearest suitable landing area." | CVR                                                                   | Crew begins mitigation actions in response to warnings.               |
| 08:37:04   | Captain reports intermittent indications and requests lower altitude.  | CVR / ATC Communication                                               | Crew recognizes worsening aircraft condition.                         |

|          |                                                      |                                                    |                                                     |
|----------|------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| 08:37:41 | Crew member: "Vibration."                            | HUMS / FDR: Rotor vibration increasing             | Mechanical instability becoming noticeable to crew. |
| 08:37:44 | Captain Mensah: "I feel it."                         | HUMS / FDR                                         | Physical vibration now perceptible in cockpit.      |
| 08:38:02 | Increased mechanical noise recorded.                 | CVR                                                | Audible evidence of worsening gearbox condition.    |
| 08:38:10 | Captain Mensah: "Torque is fluctuating."             | FDR Engine/Transmission Parameters                 | Indicates transmission instability.                 |
| 08:38:18 | Crew member: "Oil pressure dropping fast."           | HUMS / FDR / Telemetry: Rapid oil pressure decline | Confirms severe lubrication loss.                   |
| 08:38:23 | Captain Mensah: "Prepare for precautionary landing." | CVR                                                | Crew preparing for emergency landing.               |
| 08:38:54 | WARNING TONE                                         | CVR + Aircraft Warning System                      | Condition escalated beyond caution level.           |
| 08:39:01 | Captain Mensah: "We've got a serious vibration now." | HUMS / FDR: Vibration reaching peak values         | Severe mechanical instability present.              |
| 08:39:18 | Loud mechanical sound.                               | CVR                                                | Possible gearbox distress or component failure.     |
| 08:39:23 | Crew member: "RPM!"                                  | FDR Rotor Parameters                               | Concern regarding rotor speed stability.            |
| 08:39:27 | Captain Mensah: "Hold it, hold it."                  | CVR                                                | Attempt to maintain aircraft control.               |
| 08:40:02 | HIGH LEVEL WARNING TONE                              | CVR + Aircraft Warning System                      | Critical aircraft condition reached.                |
| 08:40:08 | Rapid change in rotor and engine sound.              | CVR / FDR                                          | Indicative of catastrophic mechanical failure.      |
| 08:40:22 | End of CVR recording.                                | CVR                                                | Final recorded event before loss of aircraft.       |

#### Key Correlation Findings

| Technical Event                         | Timestamp (UTC) | Artifact                      |
|-----------------------------------------|-----------------|-------------------------------|
| Gearbox temperature exceeds 100°C       | 08:31:59        | FDR / Flight Telemetry        |
| Crew reports rising gearbox temperature | 08:34:07        | CVR                           |
| Rotor vibration exceeds 1.0 g           | 08:34:09        | FDR / Flight Telemetry        |
| Oil pressure drops below 30 psi         | 08:34:34        | HUMS / FDR / Flight Telemetry |
| Crew reports vibration                  | 08:37:41        | CVR                           |
| Crew reports rapid oil pressure loss    | 08:38:18        | CVR                           |
| Crew reports severe vibration           | 08:39:01        | CVR                           |

**Observation: During the analysis of the Cockpit Voice Recorder (CVR), I observed that the flight crew progressively identified signs of a developing gearbox malfunction. The crew first reported an increase in gearbox temperature, followed by concerns regarding falling oil pressure. As the flight continued, the crew reported noticeable vibration, which later intensified and was described as serious vibration. I also observed caution and warning tones throughout the recording, indicating that the aircraft systems were detecting abnormal operating conditions. These observations were consistent with the FDR, HUMS, and telemetry data, which showed rising gearbox temperature, decreasing oil pressure, and increasing vibration leading up to the final moments of the flight. The correlation between the crew's verbal observations and the recorded flight data suggests a progressive gearbox failure prior to the crash.**

8. What communications occurred among Captain Daniel Mensah, Kwame Adu, Isaac Tetteh, Emmanuel Quartey and Linda Agyeman before the flight, and which communications are most significant to the investigation?

**Ans: My review of the phone communications revealed several exchanges between the individuals involved in the maintenance and operational release of the Bell 412.**

### **Communications Timeline Prior to Flight**

| Engineer Phone Structured Extraction                                   |                                                                                                 |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| HIVE-AV-2026-018                                                       |                                                                                                 |
| 2026-08-17T19:14:22Z - Kwame Adu to Linda Agyeman                      | The serial on the unit does not match the certificate. Stores are checking it.                  |
| 2026-08-17T19:18:40Z - Linda Agyeman to Kwame Adu                      | I see B412-OPS-883419 on the photo. Certificate says B412-OPS-883941. I am asking the supplier. |
| 2026-08-17T20:42:08Z - Kwame Adu to Isaac Tetteh                       | Ops needs the aircraft tomorrow. I will update the system after we confirm.                     |
| 2026-08-17T20:55:31Z - Emmanuel Quartey to Kwame Adu                   | Do not ignore the HUMS trend. The warning returned during the last ground run.                  |
| CREATED BY HIVE CONSULT, NANA SEI ANYEMEDU<br>PROPERTY OF HIVE CONSULT |                                                                                                 |

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$OrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unalloc (2)
      - Case\_Evidence (18)
        - 01\_Flight\_Telemetry (5)
        - 02\_HUMS (4)
        - 03\_Black\_Box\_FDR\_CVR (15)
        - 04\_GPS\_and\_Flight\_Tablet (6)
        - 05\_Weather (4)
        - 06\_ATC (6)
        - 07\_Mobile\_Evidence (7)
        - 08\_Company\_Email (6)
        - 09\_CCTV (8)
        - 10\_Crash\_Site (7)
        - 11\_Chain\_of\_Custody (10)
        - 13\_Logical\_Device\_Packages (6)
      - ProgramData (3)
        - XavierAir (3)
      - USB\_Backups (3)
      - XAL\_MAINT\_16GB (8)
      - Users (3)
      - Windows (4)
      - INF (4)
      - System32 (3)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol1/vol2/Case\_Evidence/07\_Mobile\_Evidence

Table Thumbnail Summary

Save Table as CSV

| Name                        | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(Meta) |
|-----------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-------------|
| [current folder]            |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated   |
| [parent folder]             |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated   |
| Engineer_Phone_Messages.csv |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 757  | Allocated  | unknown | Allocated   |
| Engineer_Phone_Report.html  |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 1143 | Allocated  | unknown | Allocated   |
| Pilot_Phone_Calls.csv       |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 272  | Allocated  | unknown | Allocated   |
| Pilot_Phone_Locations.csv   |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 164  | Allocated  | unknown | Allocated   |
| Pilot_Phone_Messages.csv    |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 623  | Allocated  | unknown | Allocated   |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text

timestamp\_utc,from,number,from,name,to,number,to,name,service,direction,message

2026-08-17T19:14:22Z,0287100504,Kwame Adu,0287100605,Linda Agyeman,WhatsApp,Outgoing,The serial on the unit does not match the certificate. Stores are checking it.

2026-08-17T19:18:40Z,0287100605,Linda Agyeman,0287100504,Kwame Adu,WhatsApp,Incoming,I see B412-OPS-883419 on the photo. Certificate says B412-OPS-883941. I am asking the supplier.

2026-08-17T20:42:08Z,0287100504,Kwame Adu,0287100403,Isaac Tetteh,WhatsApp,Outgoing,Ops needs the aircraft tomorrow. I will update the system after we confirm.

2026-08-17T20:55:31Z,0287100302,Emmanuel Quartey,0287100504,Kwame Adu,WhatsApp,Incoming,Do not ignore the HUMS trend. The warning returned during the last ground run.

-----METADATA-----

Content-Encoding: windows-1252

Content-Type: text/plain; charset=windows-1252

X-TIKA-Parsed-By: org.apache.tika.parser.DefaultParser

X-TIKA-Parsed-By-Full-Set: org.apache.tika.parser.DefaultParser

X-TIKA-detectedEncoding: windows-1252

X-TIKA-encodingDetector: UniversalEncodingDetector

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Sources

- The Crashed Helicopter.img\_1 Host
  - The Crashed Helicopter.img
    - vol1 (Unallocated: 0-2047)
      - \$OrphanFiles (0)
      - \$Recycle.Bin (2)
      - \$Unalloc (2)
      - Case\_Evidence (18)
        - 01\_Flight\_Telemetry (5)
        - 02\_HUMS (4)
        - 03\_Black\_Box\_FDR\_CVR (15)
        - 04\_GPS\_and\_Flight\_Tablet (6)
        - 05\_Weather (4)
        - 06\_ATC (6)
        - 07\_Mobile\_Evidence (7)
        - 08\_Company\_Email (6)
        - 09\_CCTV (8)
        - 10\_Crash\_Site (7)
        - 11\_Chain\_of\_Custody (10)
        - 13\_Logical\_Device\_Packages (6)
      - ProgramData (3)
        - XavierAir (3)
      - USB\_Backups (3)
      - XAL\_MAINT\_16GB (8)
      - Users (3)
      - Windows (4)
      - INF (4)
      - System32 (3)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Analysis Results
- OS Accounts
- Tags
- Score
- Reports

Listing

/img\_The Crashed Helicopter.img/vol1/vol2/Case\_Evidence/07\_Mobile\_Evidence

Table Thumbnail Summary

Save Table as CSV

| Name                        | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(Meta) |
|-----------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-------------|
| [current folder]            |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated   |
| [parent folder]             |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated   |
| Engineer_Phone_Messages.csv |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 757  | Allocated  | unknown | Allocated   |
| Engineer_Phone_Report.html  |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 1143 | Allocated  | unknown | Allocated   |
| Pilot_Phone_Calls.csv       |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 272  | Allocated  | unknown | Allocated   |
| Pilot_Phone_Locations.csv   |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 164  | Allocated  | unknown | Allocated   |
| Pilot_Phone_Messages.csv    |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 623  | Allocated  | unknown | Allocated   |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text

timestamp\_utc,pilot\_name,pilot\_number,other\_number,other\_name,service,direction,message,status

2026-08-18T07:20:11Z,0287100201,Captain Daniel Mensah,0287100302,Emmanuel Quartey,WhatsApp,Incoming,The gearbox warning came back during the ground run. Engineering says they are checking it. Keep an eye on the temperature indication,Read

2026-08-18T07:41:04Z,0287100201,Captain Daniel Mensah,0287100403,Isaac Tetteh,WhatsApp,Incoming,Paperwork has been cleared. Aircraft is released,Read

2026-08-18T07:45:17Z,0287100201,Captain Daniel Mensah,0287100302,Emmanuel Quartey,WhatsApp,Incoming,Copy. I will monitor the indications,Send

-----METADATA-----

Content-Encoding: windows-1252

Content-Type: text/plain; charset=windows-1252

X-TIKA-Parsed-By: org.apache.tika.parser.DefaultParser

X-TIKA-Parsed-By-Full-Set: org.apache.tika.parser.DefaultParser

X-TIKA-detectedEncoding: windows-1252

X-TIKA-encodingDetector: UniversalEncodingDetector

| Timestamp (UTC)     | Participants               | Exact Communication                                                                               |
|---------------------|----------------------------|---------------------------------------------------------------------------------------------------|
| 2026-08-17 19:14:22 | Kwame Adu to Linda Agyeman | "The serial on the unit does not match the certificate. Stores are checking it."                  |
| 2026-08-17 19:18:40 | Linda Agyeman to Kwame Adu | "I see B412-OPS-883419 on the photo. Certificate says B412-OPS-883941. I am asking the supplier." |

|                     |                                           |                                                                                                                                 |
|---------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 2026-08-17 20:42:08 | Kwame Adu to Isaac Tetteh                 | "Ops needs the aircraft tomorrow. I will update the system after we confirm."                                                   |
| 2026-08-17 20:55:31 | Emmanuel Quartey to Kwame Adu             | "Do not ignore the HUMS trend. The warning returned during the last ground run."                                                |
| 2026-08-18 07:20:11 | Emmanuel Quartey to Captain Daniel Mensah | "The gearbox warning returned during the ground run. Engineering is looking at it. Monitor the temperature indication closely." |
| 2026-08-18 07:41:04 | Isaac Tetteh to Captain Daniel Mensah     | "Paperwork is cleared. Aircraft released for flight."                                                                           |
| 2026-08-18 07:45:17 | Captain Daniel Mensah to Emmanuel Quartey | "Copy. I will monitor the indications."                                                                                         |

### **Correlation Analysis**

#### **Communication 1: Serial Number Mismatch**

Message (2026-08-17 19:14:22 UTC)

**Kwame Adu to Linda Agyeman:** "The serial on the unit does not match the certificate. Stores are checking it."

#### **Correlating Artifacts**

- Installed Component Photograph
- Component Certificate
- Supplier Invoice
- Maintenance Records

**Correlation:** This communication correlates with the installed component photograph, which displays a serial number different from the serial number recorded on the component certificate. The same discrepancy is also reflected in the supplier invoice and maintenance records. The communication therefore aligns with the documentary evidence and demonstrates that the mismatch was identified before the flight.

#### **Communication 2: Confirmation of Serial Number Discrepancy**

Message (2026-08-17 19:18:40 UTC)

**Linda Agyeman to Kwame Adu:** "I see B412-OPS-883419 on the photo. Certificate says B412-OPS-883941. I am asking the supplier."

#### **Correlating Artifacts**

- Installed Component Photograph
- Component Certificate

**Correlation:** Linda's message directly references the serial numbers observed in the photograph and certificate. Comparison of these artifacts confirms that the two serial numbers are different, supporting the accuracy of her observation.

### **Communication 3: HUMS Warning Returned**

Message (2026-08-17 20:55:31 UTC)

**Emmanuel Quartey to Kwame Adu:** "Do not ignore the HUMS trend. The warning returned during the last ground run."

#### **Correlating Artifacts**

- HUMS Dataset
- FDR Dataset
- Flight Telemetry Dataset

**Correlation:** This communication correlates with the HUMS records showing abnormal gearbox indications prior to the flight. The warning discussed in the message is further supported by the FDR and telemetry data, which later recorded rising gearbox temperature, decreasing oil pressure, and increasing vibration during flight.

### **Communication 4: Aircraft Needed for Operations**

Message (2026-08-17 20:42:08 UTC)

**Kwame Adu to Isaac Tetteh:** "Ops needs the aircraft tomorrow. I will update the system after we confirm."

#### **Correlating Artifacts**

- Maintenance Audit Log
- Maintenance Release Spreadsheet

**Correlation:** This communication occurred approximately one hour before the maintenance release status was changed from "AIRCRAFT NOT SERVICEABLE" to "CLEARED FOR FLIGHT." The timing of the message places operational requirements in close proximity to the release decision recorded in the audit logs.

### **Communication 5: Pilot Warning Before Departure**

Message (2026-08-18 07:20:11 UTC)

**Emmanuel Quartey to Captain Daniel Mensah:** "The gearbox warning returned during the ground run. Engineering is looking at it. Monitor the temperature indication closely."

#### **Correlating Artifacts**

- HUMS Dataset
- Ground Run Records

- CVR Transcript

**Correlation:** The message indicates that Captain Daniel Mensah was informed of a recurring gearbox warning before departure. This correlates with the HUMS data showing abnormal gearbox conditions and with the CVR, where the crew later discussed rising gearbox temperatures and other abnormal indications during flight.

#### **Communication 6: Aircraft Released for Flight**

Message (2026-08-18 07:41:04 UTC)

**Isaac Tetteh to Captain Daniel Mensah:** "Paperwork is cleared. Aircraft released for flight."

#### **Correlating Artifacts**

- Maintenance Audit Log
- Maintenance Release Records

**Correlation:** This communication aligns with the maintenance records showing that the aircraft had been formally released for flight. The release occurred despite earlier communications discussing the serial number discrepancy and recurring gearbox warning.

#### **Communication 7: Pilot Acknowledgement**

Message (2026-08-18 07:45:17 UTC)

**Captain Daniel Mensah to Emmanuel Quartey:** "Copy. I will monitor the indications."

#### **Correlating Artifacts**

- CVR Transcript
- FDR Data
- HUMS Data

**Correlation:** This acknowledgement correlates with the CVR, where the crew later discussed rising gearbox temperature, falling oil pressure, and increasing vibration during the flight. The communication demonstrates that the pilot was aware of the warning and actively monitoring aircraft indications.

#### **Most Significant Communications**

1. **"The serial on the unit does not match the certificate. Stores are checking it."**

First indication that the installed component documentation was questionable.

2. **"I see B412-OPS-883419 on the photo. Certificate says B412-OPS-883941."**

Independent confirmation of the mismatch.

3. **"Do not ignore the HUMS trend. The warning returned during the last ground run."**

*Direct evidence that a known warning condition existed before the flight.*

- 4. "The gearbox warning returned during the ground run. Engineering is looking at it. Monitor the temperature indication closely."**

*Demonstrates that the pilot was informed of the issue before departure.*

- 5. "Paperwork is cleared. Aircraft released for flight."**

*Critical communication linking the known concerns to the decision to operate the aircraft.*

## **Findings**

**Finding 1:** *Personnel identified a serial number discrepancy between the installed component and the supporting certificate before the flight.*

**Finding 2:** *Personnel were aware that the HUMS warning had returned during the ground run prior to departure.*

**Finding 3:** *Captain Daniel Mensah was informed of the gearbox warning before the flight and acknowledged the instruction to monitor aircraft indications.*

**Finding 4:** *Operational requirements for the aircraft were discussed before the maintenance release status was changed.*

**Finding 5:** *The aircraft was released for flight despite communications documenting unresolved maintenance concerns.*

9. Was the helicopter's route deviation itself suspicious? Compare the planned route, actual GPS/FDR track, ATC communications and weather evidence to justify your conclusion.

**Ans:** *Based on my analysis of the planned route, ATC communications, weather data, and flight evidence, the route deviation itself does not appear suspicious. The available evidence indicates that the deviation was a legitimate weather avoidance maneuver that was communicated to and approved by Air Traffic Control (ATC). However, the route deviation does not explain the subsequent mechanical deterioration observed in the aircraft systems.*

## Timeline of Events

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img\_The Crashed Helicopter.img/vol\_vo2/Case\_Evidence/06\_ATC

Table Thumbnail Summary

Save Table as CSV

| Name                               | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size    | Flags(Dir) | Known   | Flags     |
|------------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|---------|------------|---------|-----------|
| [current folder]                   |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096    | Allocated  | unknown | Allocated |
| [parent folder]                    |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096    | Allocated  | unknown | Allocated |
| ATC_Event_Timeline.csv             |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 841     | Allocated  | unknown | Allocated |
| ATC_Radio_18AUG2026.wav            |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2325898 | Allocated  | unknown | Allocated |
| ATC_Radio_18AUG2026.wav.sha256.txt |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 90      | Allocated  | unknown | Allocated |
| ATC_Transcript_18AUG2026.pdf       |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2639    | Allocated  | unknown | Allocated |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset

Text Source: File Text

| time_utc | speaker | transmission                                                                                          |
|----------|---------|-------------------------------------------------------------------------------------------------------|
| 08:27:20 | XAL-412 | Accra Control, Xavier Four-One-Two, deviating left for weather.                                       |
| 08:27:28 | ATC     | Xavier Four-One-Two, deviation approved. Report established.                                          |
| 08:32:56 | XAL-412 | Accra Control, intermittent navigation indication, position otherwise stable.                         |
| 08:33:05 | ATC     | Xavier Four-One-Two, roger, Maintain present heading and advise if assistance required.               |
| 08:37:04 | XAL-412 | Accra Control, Xavier Four-One-Two, experiencing intermittent indications and requesting lower level. |
| 08:37:15 | ATC     | Xavier Four-One-Two, descend at discretion, report level.                                             |
| 08:39:47 | XAL-412 | Accra Control, Xavier Four-One-Two—                                                                   |
| 08:39:55 | ATC     | Xavier Four-One-Two, Accra Control, say again.                                                        |
| 08:40:12 | ATC     | Xavier Four-One-Two, Accra Control, radio check.                                                      |
| 08:40:35 | ATC     | Xavier Four-One-Two, Accra Control, radio check.                                                      |

-----METADATA-----

Content-Encoding: UTF-8  
Content-Type: text/csv; charset=UTF-8; delimiter=comma  
X-TIKA-Parsed-By: org.apache.tika.parser.DefaultParser  
X-TIKA-Parsed-By-Full-Set: org.apache.tika.parser.DefaultParser  
X-TIKA-detectedEncoding: UTF-8

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img\_The Crashed Helicopter.img/vol\_vo2/Case\_Evidence/05\_Weather

Table Thumbnail Summary

Save Table as CSV

| Name                           | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags     |
|--------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-----------|
| [current folder]               |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated |
| [parent folder]                |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated |
| WEATHER BRIEFING 18AUG2026.pdf |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2574 | Allocated  | unknown | Allocated |
| Weather_Route_18AUG2026.csv    |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2585 | Allocated  | unknown | Allocated |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 75% Reset

Text Source: File Text

| timestamp_utc        | sector           | visibility_sm | wind_dir_deg | wind_speed_kt | rainfall_mm_hr | cloud_base_ft | assessment                                |
|----------------------|------------------|---------------|--------------|---------------|----------------|---------------|-------------------------------------------|
| 2026-08-18T08:02:14Z | PLANNED_CORRIDOR | 3.9           | 210          | 18.9          | 19.9           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:04:14Z | PLANNED_CORRIDOR | 4.3           | 210          | 21.3          | 15.8           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:06:14Z | PLANNED_CORRIDOR | 3.3           | 210          | 21.0          | 18.8           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:08:14Z | PLANNED_CORRIDOR | 4.1           | 210          | 21.3          | 16.4           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:10:14Z | PLANNED_CORRIDOR | 3.7           | 210          | 21.5          | 18.2           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:12:14Z | PLANNED_CORRIDOR | 4.3           | 210          | 21.1          | 17.4           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:14:14Z | PLANNED_CORRIDOR | 3.8           | 210          | 19.5          | 19.5           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:16:14Z | PLANNED_CORRIDOR | 5.0           | 210          | 18.2          | 21.6           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:18:14Z | PLANNED_CORRIDOR | 4.6           | 210          | 19.2          | 14.8           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:20:14Z | PLANNED_CORRIDOR | 3.9           | 210          | 18.4          | 21.1           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:22:14Z | PLANNED_CORRIDOR | 3.2           | 210          | 20.9          | 18.2           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:24:14Z | PLANNED_CORRIDOR | 3.5           | 210          | 19.6          | 18.5           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:26:14Z | PLANNED_CORRIDOR | 4.7           | 210          | 21.4          | 22.0           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:28:14Z | PLANNED_CORRIDOR | 5.0           | 210          | 19.7          | 13.4           | 1200          | Localized heavy rain / reduced visibility |
| 2026-08-18T08:30:14Z | REVISED_CORRIDOR | 10.1          | 220          | 15.3          | 3.2            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:32:14Z | REVISED_CORRIDOR | 10.2          | 220          | 15.7          | 4.0            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:34:14Z | REVISED_CORRIDOR | 8.4           | 220          | 13.1          | 4.8            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:36:14Z | REVISED_CORRIDOR | 9.8           | 220          | 15.6          | 2.3            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:38:14Z | REVISED_CORRIDOR | 8.3           | 220          | 14.9          | 3.8            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:40:14Z | REVISED_CORRIDOR | 9.8           | 220          | 14.6          | 3.3            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:42:14Z | REVISED_CORRIDOR | 10.9          | 220          | 15.3          | 4.3            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:44:14Z | REVISED_CORRIDOR | 11.9          | 220          | 16.8          | 4.2            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:46:14Z | REVISED_CORRIDOR | 10.9          | 220          | 14.4          | 4.5            | 2500          | Acceptable for VFR/operational deviation  |
| 2026-08-18T08:48:14Z | REVISED_CORRIDOR | 10.6          | 220          | 15.7          | 4.5            | 2500          | Acceptable for VFR/operational deviation  |

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img/The Crashed Helicopter.img/vol\_vo2/Case\_Evidence/05\_Weather

| Name                           | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(N)  |
|--------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-----------|
| [current folder]               |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated |
| [parent folder]                |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated |
| WEATHER BRIEFING_18AUG2026.pdf |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2574 | Allocated  | unknown | Allocated |
| Weather_Route_18AUG2026.csv    |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2385 | Allocated  | unknown | Allocated |

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

1 of 1 100%

WEATHER BRIEFING - XAL-412

HIVE-AV-2026-018

Summary

Localized heavy rainfall and reduced visibility were observed along the originally planned corridor west of Aconas. The revised corridor offered materially better visibility and lower rainfall.

Assessment

The route deviation is operationally reasonable based on the synthetic weather record. Weather conditions do not adequately explain the later progressive gearbox temperature increase, oil-pressure decline, or vibration trend.

Training Notice

All weather observations are synthetic and created solely for this forensic training case.

Page 1 / 1

OPERATION ROTOR TRACE - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

/img/The Crashed Helicopter.img/vol\_vo2/Case\_Evidence/04\_GPS\_and\_Flight\_Tablet

| Name                         | S | C | O | Modified Time           | Change Time         | Access Time             | Created Time            | Size | Flags(Dir) | Known   | Flags(N)  |
|------------------------------|---|---|---|-------------------------|---------------------|-------------------------|-------------------------|------|------------|---------|-----------|
| [current folder]             |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated |
| [parent folder]              |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 4096 | Allocated  | unknown | Allocated |
| ACC_TARKWA_FLIGHT_PLAN.pdf   |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 2399 | Allocated  | unknown | Allocated |
| Actual_Flight_18AUG2026.gpx  |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 6339 | Allocated  | unknown | Allocated |
| Actual_Flight_18AUG2026.kml  |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 1861 | Allocated  | unknown | Allocated |
| Planned_Route_ACC_TARKWA.gpx |   |   |   | 2026-08-18 13:00:00 UTC | 0000-00-00 00:00:00 | 2026-08-18 00:00:00 UTC | 2026-08-18 13:00:00 UTC | 5421 | Allocated  | unknown | Allocated |

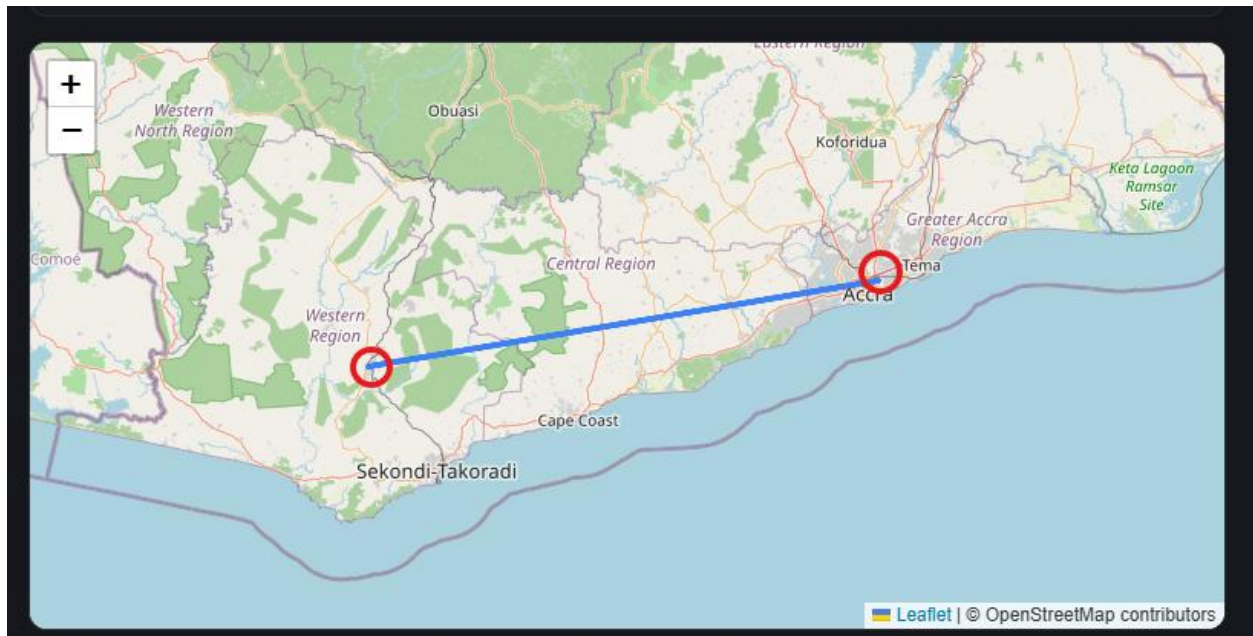
Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Go to Page: Script: Latin - Basic

```
<?xml version="1.0" encoding="UTF-8"?>
<gpx version="1.1" creator="HIVE CONSULT" xmlns="http://www.topografix.com/GPX/1/1">
  <tr><name>Operation Rotor Trace</name><id><tr>
    <trkpt lat="5.6052" lon="-0.1668">
      <trkpt lat="5.601336708860759" lon="-0.18994177215189872">
        <trkpt lat="5.597473417721519" lon="-0.21308354430379747">
          <trkpt lat="5.593610126582279" lon="-0.23622531645569602">
            <trkpt lat="5.589746835443038" lon="-0.25936708860759494">
              <trkpt lat="5.5858835443037979" lon="-0.28250886075949366">
                <trkpt lat="5.58202023164557" lon="-0.305606329113904">
                  <trkpt lat="5.57815696202316" lon="-0.32070246596329116">
                    <trkpt lat="5.5742936708860755" lon="-0.3519341772151899">
                      <trkpt lat="5.570430379746836" lon="-0.375079493670886">
                        <trkpt lat="5.566567088607595" lon="-0.3982177215189874">
                          <trkpt lat="5.562703797468354" lon="-0.4213594936708861">
                            <trkpt lat="5.558840506329114" lon="-0.4445012658227848">
                              <trkpt lat="5.554977215189873" lon="-0.46764303797468354">
                                <trkpt lat="5.551113924050632" lon="-0.4907848101265823">
                                  <trkpt lat="5.547250632911393" lon="-0.5139263822784811">
                                    <trkpt lat="5.543387341772152" lon="-0.5370683544303798">
                                      <trkpt lat="5.539524050632911" lon="-0.5602101265822785">
                                        <trkpt lat="5.535660759493671" lon="-0.5833518987341773">
                                          <trkpt lat="5.53179746835443" lon="-0.606493670886076">
```



| <b>Time (UTC)</b> | <b>Artifact</b>             | <b>Event</b>                                                                                        |
|-------------------|-----------------------------|-----------------------------------------------------------------------------------------------------|
| 08:26:14          | Weather_Route_18AUG2026.csv | Planned corridor experiencing heavy rainfall (22.0 mm/hr) and reduced visibility (4.7 km).          |
| 08:27:20          | ATC_Event_Timeline.csv      | XAL-412 reported: "Accra Control, Xavier Four-One-Two, deviating left for weather."                 |
| 08:27:28          | ATC_Event_Timeline.csv      | ATC responded: "Xavier Four-One-Two, deviation approved. Report established."                       |
| 08:30:14          | Weather_Route_18AUG2026.csv | Revised corridor showed improved visibility (10.1 km) and significantly lower rainfall (3.2 mm/hr). |
| 08:32:56          | ATC_Event_Timeline.csv      | Pilot reported intermittent navigation indications but stated position remained stable.             |
| 08:37:04          | ATC_Event_Timeline.csv      | Pilot reported intermittent indications and requested a lower level.                                |
| 08:39:47          | ATC_Event_Timeline.csv      | Final incomplete transmission from XAL-412.                                                         |

### Correlation of Artifacts

- 1. Planned Route vs Weather Evidence:** The weather data shows that the planned corridor was affected by localized heavy rainfall and poor visibility.

**Planned Corridor Conditions**

| <b>Time</b>          | <b>Visibility</b> | <b>Rainfall</b>   |
|----------------------|-------------------|-------------------|
| 08:26:14 UTC         | 4.7 km            | 22.0 mm/hr        |
| Earlier observations | 3.2 - 5.0 km      | 14.8 - 21.6 mm/hr |

**Revised Corridor Conditions**

| <b>Time</b>  | <b>Visibility</b> | <b>Rainfall</b> |
|--------------|-------------------|-----------------|
| 08:30:14 UTC | 10.1 km           | 3.2 mm/hr       |
| 08:32:14 UTC | 10.2 km           | 4.0 mm/hr       |
| 08:36:14 UTC | 9.8 km            | 2.3 mm/hr       |

**Observation:** The deviation reported at 08:27:20 UTC occurred while the planned route was experiencing significantly poorer visibility and substantially heavier rainfall than the revised corridor. The weather data supports the pilot's statement that the deviation was made for weather avoidance.

**2. ATC Communications vs Route Deviation**

**Communication Evidence**

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| 08:27:20 UTC | "Accra Control, Xavier Four-One-Two, deviating left for weather." |
| 08:27:28 UTC | "Xavier Four-One-Two, deviation approved. Report established."    |

The pilot explicitly informed ATC of the reason for the deviation and received approval eight seconds later. This demonstrates that the route change was neither concealed nor unauthorized.

**3. Weather Evidence vs Official Weather Assessment**

**The weather briefing states:** "Localized heavy rainfall and reduced visibility were observed along the originally planned corridor west of Accra."

**It further stated:** "The revised corridor offered materially better visibility and lower rainfall."

The weather briefing independently confirms what is observed in the weather-route dataset. Both sources show that the revised route offered better flying conditions.

**Corroborating Evidence**

| <b>ATC Communications</b>                                                              | <b>Weather_Route_18 AUG2026.csv</b>                                                                         | <b>WEATHER BRIEFING_18AU G2026.pdf</b>                                                                             | <b>Planned Route (GPX)</b>                                                                             |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <i>Corroborate that the pilot intentionally deviated for weather and notified ATC.</i> | <i>Corroborates that the planned corridor experienced poorer weather conditions than the revised route.</i> | <i>Corroborates that the revised corridor offered better visibility and lower rainfall than the planned route.</i> | <i>Provides the intended route used as the baseline for comparison against the actual flight path.</i> |

## **Findings**

**Finding 1:** At 08:27:20 UTC, the pilot informed ATC that the helicopter was deviating left due to weather.

**Finding 2:** At 08:27:28 UTC, ATC approved the deviation, indicating that the maneuver was authorized.

**Finding 3:** Weather data recorded heavy rainfall and reduced visibility along the planned route immediately before the deviation.

**Finding 4:** The revised corridor provided substantially better visibility and lower rainfall, making the deviation operationally reasonable.

**Finding 5:** No evidence was identified indicating that the route deviation was unauthorized, concealed, or inconsistent with the prevailing weather conditions.

**Investigative Assessment:** The route deviation itself is not suspicious. The ATC communications, weather-route data, and weather briefing all support the conclusion that the deviation was made to avoid adverse weather conditions and was approved by air traffic control. While the helicopter later experienced significant gearbox abnormalities, the available evidence indicates that the route deviation was a legitimate operational decision rather than an indicator of wrongdoing or an attempt to conceal activity. The weather briefing specifically notes that the route deviation was operationally reasonable and does not explain the later gearbox temperature increase, oil pressure decline, or vibration trend.

- Based on all available digital evidence, construct a final timeline from the maintenance work three days before the crash through the final recorded flight event, and provide your forensic assessment of the most significant factors that contributed to the incident.

**Ans:** Based on the totality of the digital evidence, the incident was preceded by known maintenance concerns, including a recurring gearbox warning and a component documentation discrepancy. Despite these issues, the aircraft was released for flight. During the flight, the helicopter experienced a progressive gearbox failure characterized by rising gearbox temperature, falling oil pressure, increasing vibration, and severe mechanical instability. The HUMS, FDR, telemetry, ATC, CVR, maintenance, communication, browser history, and audit log evidence collectively support the conclusion that unresolved maintenance issues existed before departure and that the gearbox failure was the primary factor contributing to the accident.

## Final Timeline of Events

### a. 15 August 2026 - Maintenance Activities (3 Days Before Crash)

| Date/Time (UTC) | Event                                                                                     | Artifact              |
|-----------------|-------------------------------------------------------------------------------------------|-----------------------|
| 2026-08-15      | Gearbox component replacement performed on Bell 412 XAL-412.                              | Maintenance Records   |
| 2026-08-15      | Replacement component installed with serial number recorded in maintenance documentation. | Maintenance Records   |
| 2026-08-15      | Component certificate and supporting documentation associated with replacement unit.      | Component Certificate |

**Finding:** The maintenance activity introduced the component later found to have a serial number discrepancy between the installed unit and supporting documentation.

### b. 17 August 2026 - Discovery of Documentation Issues

| Time (UTC) | Event                                                                                       | Artifact       |
|------------|---------------------------------------------------------------------------------------------|----------------|
| 19:14:22   | Kwame Adu: "The serial on the unit does not match the certificate. Stores are checking it." | Phone Messages |
| 19:18:40   | Linda Agyeman confirms certificate and installed serial numbers differ.                     | Phone Messages |

**Finding:** Personnel identified a serial number discrepancy before the flight.

### c. 17 August 2026 - Warning Signs Prior to Release

| Time (UTC) | Event                                                                                              | Artifact       |
|------------|----------------------------------------------------------------------------------------------------|----------------|
| 20:42:08   | Kwame Adu: "Ops needs the aircraft tomorrow."                                                      | Phone Messages |
| 20:55:31   | Emmanuel Quartey: "Do not ignore the HUMS trend. The warning returned during the last ground run." | Phone Messages |

**Finding:** Maintenance personnel were aware of a recurring gearbox warning before the aircraft was released.

### d. 17 August 2026 - Maintenance Status Modification

| Time (UTC) | Event                                                                                                      | Artifact              |
|------------|------------------------------------------------------------------------------------------------------------|-----------------------|
| 21:43:49   | Maintenance status changed from <b>AIRCRAFT NOT SERVICEABLE</b> to <b>CLEARED FOR FLIGHT</b> by Kwame Adu. | Maintenance Audit Log |

**Finding:** The aircraft was released for service despite documented concerns regarding component documentation and recurring warnings.

**e. 17 August 2026 - Post-Modification Browser Activity**

| <b>Time (UTC)</b> | <b>Event</b>                                | <b>Artifact</b> |
|-------------------|---------------------------------------------|-----------------|
| 21:58:07          | Search: "remove recent files windows 11"    | Browser History |
| 22:01:41          | Search: "clear browser history permanently" | Browser History |

**Finding: The timing of these searches is consistent with attempts to remove or conceal activity occurring shortly after the maintenance release record was changed.**

**f. 18 August 2026 - Pre-Flight Communications**

| <b>Time (UTC)</b> | <b>Event</b>                                                                                            | <b>Artifact</b> |
|-------------------|---------------------------------------------------------------------------------------------------------|-----------------|
| 07:20:11          | Emmanuel Quartey advises Captain Daniel Mensah that the gearbox warning returned during the ground run. | Pilot Messages  |
| 07:41:04          | Isaac Tetteh informs Captain Mensah that paperwork is cleared and aircraft released for flight.         | Pilot Messages  |
| 07:45:17          | Captain Mensah acknowledges and states he will monitor indications.                                     | Pilot Messages  |

**Finding: The pilot was informed of the warning condition before departure.**

**g. 18 August 2026 - Route Deviation**

| <b>Time (UTC)</b> | <b>Event</b>                              | <b>Artifact</b>    |
|-------------------|-------------------------------------------|--------------------|
| 08:27:20          | Pilot reports deviation left for weather. | ATC Communications |
| 08:27:28          | ATC approves deviation.                   | ATC Communications |

**Finding: The route deviation was justified by adverse weather and was not suspicious.**

**h. 18 August 2026 - Development of Mechanical Failure**

| <b>Time (UTC)</b> | <b>Event</b>                       | <b>Artifact</b>        |
|-------------------|------------------------------------|------------------------|
| 08:31:59          | Gearbox temperature exceeds 100°C. | FDR / Telemetry        |
| 08:34:09          | Vibration exceeds 1.0 g.           | FDR / Telemetry        |
| 08:34:34          | Oil pressure drops below 30 psi.   | HUMS / FDR / Telemetry |

**Finding: The first abnormal condition was gearbox overheating, followed by increasing vibration and declining oil pressure.**

**i. 18 August 2026 - Crew Recognition of Failure**

| <b>Time (UTC)</b> | <b>Event</b> | <b>Artifact</b> |
|-------------------|--------------|-----------------|
|-------------------|--------------|-----------------|

|          |                                              |     |
|----------|----------------------------------------------|-----|
| 08:34:07 | Crew reports gearbox temperature increasing. | CVR |
| 08:35:32 | Crew reports oil pressure moving down.       | CVR |
| 08:37:41 | Crew reports vibration.                      | CVR |
| 08:38:18 | Crew reports oil pressure dropping fast.     | CVR |
| 08:39:01 | Crew reports serious vibration.              | CVR |

*Finding: The crew's observations matched the technical evidence recorded by the aircraft systems.*

**j. 18 August 2026 - Final Recorded Events**

| <b>Time (UTC)</b> | <b>Event</b>                            | <b>Artifact</b> |
|-------------------|-----------------------------------------|-----------------|
| 08:39:18          | Loud mechanical sound recorded.         | CVR             |
| 08:39:23          | Crew member calls out "RPM!"            | CVR             |
| 08:40:02          | High-level warning tone recorded.       | CVR             |
| 08:40:08          | Rapid change in rotor and engine sound. | CVR             |
| 08:40:22          | End of CVR recording.                   | CVR             |

## Forensic Assessment

Based on the available digital evidence, I identified four significant factors that contributed to the incident.

**1. Known Gearbox Warning Prior to Flight:** Communications between Emmanuel Quartey, Kwame Adu, and Captain Daniel Mensah demonstrate that personnel were aware of a recurring HUMS warning before the aircraft departed. This finding is corroborated by HUMS records, FDR data, telemetry data, and CVR observations.

**2. Component Documentation Discrepancy:** The communications between Kwame Adu and Linda Agyeman establish that a serial number discrepancy existed between the installed component and supporting documentation. This finding is corroborated by the installed component photographs, component certificate, supplier invoice, and maintenance records.

**3. Release of Aircraft Despite Known Concerns:** Maintenance audit logs show that the aircraft status was changed from AIRCRAFT NOT SERVICEABLE to CLEARED FOR FLIGHT at 21:43:49 UTC on 17 August 2026. This occurred after discussions regarding the serial number discrepancy and recurring gearbox warning.

**4. Progressive Gearbox Failure During Flight:** HUMS, FDR, telemetry, and CVR evidence consistently show a progression from gearbox overheating to oil pressure loss, increasing vibration, mechanical instability, and eventual loss of control. The sequence was independently recorded across multiple systems and corroborated by crew observations.

## Limitations

1. *The investigation was conducted using the digital artifacts provided as part of the case scenario. No physical examination of the Bell 412 helicopter, gearbox components, or crash site was performed.*
2. *The findings are based on the available maintenance records, communications, audit logs, browser history, USB artifacts, HUMS data, FDR data, telemetry records, ATC communications, weather records, and CVR transcript. Additional evidence not included in the dataset may provide further context.*
3. *User attribution was performed using account information, audit logs, timestamps, and system artifacts. While these artifacts associate actions with specific accounts, they do not conclusively prove who was physically operating the device at the time.*
4. *The serial number discrepancy identified during the investigation was assessed using the supplied documentation and communications. Independent verification of component authenticity was outside the scope of the available evidence.*
5. *The conclusions are limited to the evidence contained within the fictional case dataset and should be interpreted within the context of the exercise.*

## **Recommendations**

1. *Maintenance systems should require dual authorization before changing an aircraft status from **"AIRCRAFT NOT SERVICEABLE"** to **"CLEARED FOR FLIGHT"**.*
2. *All recurring HUMS warnings should be fully investigated and resolved before an aircraft is released for operational use.*
3. *Component serial numbers, certificates, invoices, and maintenance records should be independently verified before installation and release approval.*
4. *Maintenance audit logs should be protected using tamper-resistant logging and continuous monitoring to improve accountability.*
5. *Organizations should implement procedures to review unusual user activity, including searches related to deleting logs, removing metadata, or clearing system history following maintenance actions.*
6. *Flight crews should receive complete documentation of unresolved maintenance concerns before accepting an aircraft for operation.*
7. *Regular audits of maintenance documentation and release procedures should be conducted to identify discrepancies before they affect flight operations.*

## **Examiner's Opinion**

*Based on the available digital evidence, the most significant contributing factors in this fictional incident were the known gearbox warning condition, the component documentation discrepancy, and the decision to release the aircraft for flight despite unresolved concerns. The HUMS, FDR, telemetry, CVR, communication records, and maintenance artifacts collectively indicate that warning signs existed before departure and were later reflected in the progressive gearbox failure observed during the final stages of flight.*